

Inverse q -Analogues

**Branch Geometry, Asymptotic Inversion, and Computation for
 q -Pochhammer Symbols, Gaussian Coefficients, and Related Functions**

A research companion to the ProveIt q -series monograph

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Abstract

This report develops a systematic theory of inverse functions for the principal objects of elementary and basic hypergeometric analysis. The central point is that “the inverse” is almost never a single globally defined function. One must specify the parameter being recovered, the real or complex domain, a branch, and the singular set across which continuation fails. We therefore study partial inverses with respect to the argument, the base q , the order, and the discrete or continuously interpolated parameters.

For finite q -Pochhammer products we give a complete real branch classification in the argument, compute the exact discriminant at level zero, and introduce a *secondary discriminant* that detects collisions of critical values. The first nontrivial positive-base collision occurs already for $(a; q)_5$: an exact reciprocal degree-ten polynomial has a unique root $q_0 \in (0, 1)$, numerically

$$q_0 = 0.8401533783083766586629364318891329431 \dots$$

At this base two distinct negative critical values coincide. This creates a Maxwell-type locus in the inverse covering and shows why naive global ordering conjectures fail.

For the infinite product $(a; q)_\infty$, we define and analyze its principal real inverse $\mathcal{A}_q : (0, \infty) \rightarrow (-\infty, 1)$, derive endpoint series, a log-periodically corrected large-target asymptotic, and the asymptotic location and height of every nonprincipal real branch point. Inversion with respect to the order is reduced exactly to $\mathcal{A}_q$. Inversion with respect to the base is proved globally monotone in the principal sign regimes and is expanded both at $q = 0$ and $q = 1$, the latter in polylogarithmic form.

For Gaussian coefficients, positivity of coefficients yields a global theorem: for every nontrivial $\begin{bmatrix} n \\ k \end{bmatrix}_q$ and every $y > 1$, there is exactly one positive real base q with $\begin{bmatrix} n \\ k \end{bmatrix}_q = y$. Bernoulli expansions are then reverted to all orders near $q = 1$, and reciprocal symmetry controls the large- q inverse. Continuous inverses in the upper and lower parameters are developed through the q -gamma representation. Separate chapters treat inverse q -numbers, q -factorials, q -gamma, q -digamma, q -beta, Euler q -exponentials, basic hypergeometric series, theta functions, and broader positive-coefficient q -analogs.

The report distinguishes proved statements, reproducible symbolic or numerical findings, and conjectures. The accompanying Python program regenerates all figures and data, including high-precision branch-point calculations and exact secondary discriminants through order six.

Relation to the source monograph and status of claims

The source document is Vladimir Reshetnikov’s proof-oriented monograph *q -Pochhammer Symbols and q -Binomial Coefficients* in the ProveIt repository [1]. It supplies the forward theory used here: finite and infinite products, logarithmic derivatives, complex orders, Gaussian coefficients, q -gamma and q -beta functions, and Bernoulli asymptotics. The present report is a companion organized around inversion, branch geometry, conditioning, and algorithms rather than a repetition of that forward theory.

A label at the beginning of a statement has the following meaning.

PAPER PROOF	a proof is supplied here, or a standard cited theorem is used explicitly;
SYMBOLIC/NUMERICAL	exact symbolic algebra or high-precision numerics are reproducible from the archive;
CONJECTURAL	evidence and a precise formulation are given, but no proof is claimed.

Here and below, “paper proof” means proved in this report or imported from a cited mathematical source; it does *not* assert Lean formalization. The live Lean corpus supplies selected finite Gaussian/Pochhammer algebra, including `Fabius.finiteQPochhammerIn`, `Fabius.gauss`

ianBinomial, and `Fabius.finite_qBinomial_theorem`, together with real and complex infinite-product convergence such as `Fabius.tendsto_finiteQPochhammerIn` and `Fabius.tendsto_finiteQPochhammerIn_complex`. The exact base-reversal crosswalk `Fabius.finiteQPochhammerIn_base_reversal` requires $a \neq 0$ and $q \neq 0$. None of this report’s specialized inverse, critical-atlas, discriminant/Maxwell, monodromy, branch-geometry, or inverse asymptotic theorems currently has an exact public Lean counterpart. The source monograph’s 184-row formalization ledger records 35 exact, 24 partial, 122 without counterparts, and 3 interface-only rows; in particular, evaluated Gaussian positivity and infinite-product convergence do not establish the coefficientwise, derivative-sign, inverse, or asymptotic claims made here.

Some results, especially the secondary-discriminant calculations and the asymptotic organization of the nonprincipal inverse branches, appear not to be standard in this form. No assertion of bibliographic priority is made without an exhaustive literature search. Numerical experiments are not substitutes for interval or formal proof; the exact polynomial identities, however, can be checked in a computer algebra system over $\mathbb{Z}[q]$.

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Part I

Foundations of parameter-wise inversion

1 What it means to invert a q -analogue

1.1 Partial inverses, not one universal inverse

Let

$$F(p_1, \dots, p_m) = y$$

be a scalar q -analogue with several parameters. One scalar equation does not determine all m parameters: generically its solution set is an $(m-1)$ -dimensional level hypersurface. Thus throughout this report an *inverse with respect to a parameter* means that all other parameters are fixed and one chosen parameter is recovered. A simultaneous inverse requires an equal number of independent equations or additional normalization conditions.

Definition 1.1 (Partial inverse branch). Fix auxiliary parameters λ and write $F(p; \lambda)$. A partial inverse branch is a function $p = G(y; \lambda)$ on a domain U such that

$$F(G(y; \lambda); \lambda) = y,$$

with a specified initial value, real interval, or analytic-continuation path. Its branch locus consists of singular values reached from points where $F_p = 0$, together with values caused by poles or singular auxiliary parameters.

The main objects and their inversion variables are summarized in [Table 1.1](#).

Table 1.1: Inverse variables and principal tools.

Object	Recoverable parameter	Principal mechanism
$(a; q)_n, (a; q)_\infty$	argument a	algebraic covering for finite n ; entire-function covering for $n = \infty$
	base q	monotonicity on real sign domains; polynomial or analytic branch continuation
	order n or α	discrete recurrence or complex-order quotient
$\begin{bmatrix} n \\ k \end{bmatrix}_q$	base q	nonnegative coefficients, Bernoulli reversion, reciprocal symmetry
	n, k or α, β	ratio recurrences and q -gamma interpolation
$[x]_q, [n]_q!$	x, n, q	elementary logarithm or positive-polynomial inversion
$\Gamma_q(x)$	x	two real branches separated by the unique minimum
	q	known monotonicity sectors and endpoint asymptotics
$B_q(a, b)$	a, b	strict coordinate monotonicity
	q	Lambert-series derivative; complete integer-parameter theorem
Euler q -exponentials	argument	exact reduction to the inverse infinite product
Basic hypergeometric and theta functions	argument and shape parameters	local Lagrange inversion, critical values, quasi-periodicity

1.2 Real inverse, complex inverse, and Riemann surface

On a real interval, a continuous strictly monotone function has one ordinary inverse. In the complex plane, even a polynomial of degree two has two sheets. A useful global object is therefore the analytic curve

$$\mathcal{R}_F = \{(p, y) \in \mathbb{C}^2 : F(p) = y\},$$

projected to the y -plane. Away from critical values this projection is a covering map. Analytic continuation along loops permutes the sheets; this is the monodromy action. For finite q -Pochhammer products the covering is algebraic, whereas for the infinite product it has infinitely many sheets and branch points.

1.3 Normalization and branch labels

A branch can be labelled in several non-equivalent ways:

- (i) by the real interval containing the recovered parameter;
- (ii) by a base point (p_0, y_0) and analytic continuation from it;
- (iii) by an asymptotic condition, such as $p(y) \rightarrow 1$ as $y \rightarrow 0^+$;
- (iv) by a logarithm index or winding number;
- (v) by ordering roots at a chosen regular target.

The report uses interval labels for real branches and continuation labels for complex branches. A numerical routine that omits such a label is incomplete: Newton's method can jump between sheets.

2 Local analytic inversion and singularity theory

2.1 The inverse function theorem and sensitivities

Theorem 2.1 (PAPER PROOF Local partial inverse). *Let $F(p, \lambda)$ be holomorphic near (p_0, λ_0) , put $y_0 = F(p_0, \lambda_0)$, and assume $F_p(p_0, \lambda_0) \neq 0$. Then there is a unique holomorphic branch $p = G(y, \lambda)$ near (y_0, λ_0) satisfying $F(G(y, \lambda), \lambda) = y$ and $G(y_0, \lambda_0) = p_0$. Its first sensitivities are*

$$G_y = \frac{1}{F_p}, \quad G_{\lambda_j} = -\frac{F_{\lambda_j}}{F_p},$$

where the derivatives on the right are evaluated at $p = G(y, \lambda)$.

Proof. The holomorphic implicit-function theorem gives the branch. Differentiating $F(G(y, \lambda), \lambda) = y$, first in y and then in λ_j , gives the displayed formulas. $\square$

Corollary 2.2 (PAPER PROOF First inverse derivatives). *For one variable,*

$$\begin{aligned} G'(y) &= \frac{1}{F'(p)}, \\ G''(y) &= -\frac{F''(p)}{F'(p)^3}, \\ G'''(y) &= \frac{3F''(p)^2 - F'(p)F'''(p)}{F'(p)^5}. \end{aligned}$$

A dimensionless relative condition number is

$$\kappa_{p \leftarrow y} = \left| \frac{y}{pF_p(p)} \right|.$$

At $p = 0$, or when relative error is inappropriate, the absolute condition number $1/|F_p|$ is preferable. Divergence of either quantity near $F_p = 0$ is not a numerical artifact: it is the analytic signature of a branch point.

2.2 Lagrange–Buerhmann coefficients

Theorem 2.3 (PAPER PROOF Lagrange–Buerhmann inversion). *Let F be holomorphic near p_0 , with $F'(p_0) \neq 0$, and set $y_0 = F(p_0)$. The local inverse is*

$$G(y) = p_0 + \sum_{m \geq 1} \frac{(y - y_0)^m}{m!} \frac{d^{m-1}}{dp^{m-1}} \left(\frac{p - p_0}{F(p) - F(p_0)} \right)^m \Big|_{p=p_0}.$$

The radius of convergence is bounded by the nearest singular value of the chosen sheet.

Proof. Apply the classical Lagrange inversion theorem to $w = p - p_0$ and $z = y - y_0 = w\phi(w)$, where $\phi(0) = F'(p_0) \neq 0$. The coefficient formula follows after rewriting $1/\phi(w) = (p - p_0)/(F(p) - F(p_0))$. Analytic continuation of the inverse can fail only when the direct map or its local inverse becomes singular, hence a critical or singular value bounds the disk. $\square$

If

$$F(p_0 + w) = y_0 + f_1 w + f_2 w^2 + f_3 w^3 + \cdots$$

and

$$G(y_0 + z) = p_0 + g_1 z + g_2 z^2 + g_3 z^3 + \cdots,$$

then

$$g_1 = \frac{1}{f_1}, \quad g_2 = -\frac{f_2}{f_1^3}, \quad g_3 = \frac{2f_2^2}{f_1^5} - \frac{f_3}{f_1^4}.$$

This three-term template recurs throughout the report.

2.3 Simple branch points and Puiseux inversion

Theorem 2.4 (PAPER PROOF Universal square-root law). *Suppose*

$$F'(p_c) = 0, \quad F''(p_c) \neq 0, \quad y_c = F(p_c).$$

Then the two local inverse sheets have Puiseux expansions

$$p = p_c \pm \sqrt{\frac{2(y - y_c)}{F''(p_c)}} - \frac{F'''(p_c)}{3F''(p_c)^2}(y - y_c) + O((y - y_c)^{3/2}),$$

with a consistent choice of square root.

Proof. Write $u = p - p_c$, $v = y - y_c$, and expand

$$v = \frac{F''(p_c)}{2}u^2 + \frac{F'''(p_c)}{6}u^3 + O(u^4).$$

Insert $u = c_1 v^{1/2} + c_2 v + O(v^{3/2})$. Comparing the coefficients of v and $v^{3/2}$ gives $c_1^2 = 2/F''(p_c)$ and $c_2 = -F'''(p_c)/(3F''(p_c)^2)$. $\square$

More generally, if the first nonzero derivative is $F^{(r)}(p_c)$, the local inverse has r -th-root ramification. This provides the correct local model for multiple critical points and for coalescing branches.

2.4 Discriminants, caustics, and Maxwell sets

For a polynomial $F(p; \lambda)$, the critical-value polynomial is

$$\Delta(y; \lambda) = \text{Disc}_p(F(p; \lambda) - y).$$

Its roots in y are the critical values, counted with multiplicity. Two different degeneracies should be distinguished.

Definition 2.5 (Caustic and Maxwell locus). The *caustic* in parameter space is the set where a critical point itself becomes multiple, so that $F_p = F_{pp} = 0$. The *Maxwell locus* is the set where two distinct simple critical points have the same critical value. For a one-parameter polynomial family both are detected, with multiplicity, by

$$\mathfrak{M}(\lambda) = \text{Disc}_y \Delta(y; \lambda).$$

The terminology is useful because the inverse geometry is different. At a caustic, local ramification order changes. At a Maxwell point, two ordinary square-root branch points project to the same target value.

2.5 Several unknown parameters

If $F : \mathbb{C}^m \rightarrow \mathbb{C}^m$, a simultaneous local inverse exists when $\det DF \neq 0$. Its differential is

$$d\mathbf{p} = (DF)^{-1} d\mathbf{y}.$$

For a scalar $F : \mathbb{C}^m \rightarrow \mathbb{C}$, no such matrix inverse exists; the tangent space of the level set is the nullspace of ∇F . This observation prevents a common ambiguity in phrases such as “invert with respect to q and all parameters”: the mathematically meaningful program is to construct every one-parameter partial inverse and then study coupled systems separately.

3 Derivative kernels for q -products

3.1 Finite and infinite argument derivatives

For $N \in \mathbb{N} \cup \{\infty\}$, write

$$P_N(a; q) = (a; q)_N.$$

For $N = \infty$ assume $0 < |q| < 1$. Define

$$S_{r,N}(a; q) = \sum_{j=0}^{N-1} \frac{q^{rj}}{(1 - aq^j)^r},$$

with the evident infinite interpretation.

Proposition 3.1 (PAPER PROOF Logarithmic derivative hierarchy). *Away from the zeros of P_N , for every $r \geq 1$,*

$$\frac{\partial^r}{\partial a^r} \log P_N(a; q) = -(r-1)! S_{r,N}(a; q).$$

In particular,

$$P'_N(a; q) = -P_N(a; q) S_{1,N}(a; q)$$

and

$$\frac{P''_N(a; q)}{P_N(a; q)} = S_{1,N}(a; q)^2 - S_{2,N}(a; q).$$

Proof. Differentiate $\sum_j \log(1 - aq^j)$ term by term. In the infinite case the series and all its derivatives converge locally uniformly away from the zero set. The final identities follow from differentiating an exponential. $\square$

On $a < 1$ with $0 < q < 1$, all factors are positive and

$$(-1)^r P_N^{(r)}(a; q) > 0$$

for every available derivative $1 \leq r \leq N$, and for every $r \geq 1$ when $N = \infty$. For finite N , this follows by differentiating the product and summing over r -element subsets; passage to the infinite product is by locally uniform convergence.

3.2 Base derivatives

Proposition 3.2 (PAPER PROOF Derivative with respect to the base). *For finite N ,*

$$\frac{\partial}{\partial q} \log(a; q)_N = -a \sum_{j=1}^{N-1} \frac{j q^{j-1}}{1 - aq^j}.$$

For $N = \infty$ and $|q| < 1$, the same formula holds with the sum extended to infinity, locally uniformly away from zeros.

Proof. The $j = 0$ factor is independent of q . Differentiate every other factor. The infinite derivative series converges normally on compact subsets on which the denominators stay separated from zero. $\square$

This kernel immediately settles the principal real sign regimes. If $0 < a < 1$, the derivative is negative; if $a < 0$, it is positive. When $a > 1$, denominators change sign as q crosses the real zeros $q = a^{-1/j}$, and a branch-by-branch analysis is unavoidable.

3.3 Complex order derivatives

For a fixed branch of q^α , define

$$(a; q)_\alpha = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}.$$

Put $b = aq^\alpha$. Then

$$\begin{aligned} \frac{\partial}{\partial \alpha} \log(a; q)_\alpha &= (\log q) \sum_{j \geq 0} \frac{bq^j}{1 - bq^j}, \\ \frac{\partial^2}{\partial \alpha^2} \log(a; q)_\alpha &= (\log q)^2 \sum_{j \geq 0} \frac{bq^j}{(1 - bq^j)^2}. \end{aligned}$$

For real $0 < q < 1$ and $0 < b < 1$, the first quantity is negative and the second positive. Thus the order dependence is strictly decreasing and log-convex on that real branch.

3.4 Truncation bounds for certified evaluation

Lemma 3.3 ([PAPER PROOF](#) Infinite-product tail bound). *Let $0 < |q| < 1$, choose N so that $\rho = |a| |q|^N < 1$, and write $T_N = (aq^N; q)_\infty$. For the logarithm obtained by continuation from $a = 0$,*

$$|\log T_N| \leq \frac{\rho}{(1 - |q|)(1 - \rho)}.$$

Consequently,

$$|T_N - 1| \leq \exp\left(\frac{\rho}{(1 - |q|)(1 - \rho)}\right) - 1.$$

Proof. For $|z| < 1$, $|\log(1 - z)| \leq |z|/(1 - |z|)$. Sum this estimate for $z = aq^j$, $j \geq N$, and bound every denominator below by $1 - \rho$. The second estimate follows from $T_N = \exp(\log T_N)$. $\square$

These bounds give a direct route to interval or ball arithmetic. They also explain why evaluation in the logarithmic domain is preferable near $q = 1$, where direct multiplication underflows long before the mathematical product is zero.

Part II

Inverse q -Pochhammer theory

4 Finite products: inversion in the argument

4.1 The real branch decomposition

Fix $n \geq 1$ and $0 < q < 1$, and put

$$P_n(a) = \prod_{j=0}^{n-1} (1 - aq^j).$$

Its zeros

$$r_j = q^{-j}, \quad 0 \leq j \leq n-1,$$

are real, simple, and strictly increasing.

Theorem 4.1 (PAPER PROOF Complete real critical-point structure). *For $n \geq 2$, the following statements hold.*

- (i) *On $(-\infty, 1)$, P_n is positive and strictly decreasing from $+\infty$ to 0. Hence every $y > 0$ has exactly one inverse there.*
- (ii) *For each $k = 0, \dots, n-2$, the interval*

$$I_k = (q^{-k}, q^{-(k+1)})$$

contains exactly one critical point $c_{n,k}(q)$. It is simple. The sign of P_n on I_k is $(-1)^{k+1}$, and $P_n(c_{n,k})$ is the unique extremum on that interval.

- (iii) *On the last interval $(q^{-(n-1)}, \infty)$, there is no critical point. The map is monotone from 0 to $+\infty$ if n is even and from 0 to $-\infty$ if n is odd.*

Proof. On $a < 1$, every factor is positive and

$$\frac{P'_n(a)}{P_n(a)} = -S_{1,n}(a; q) < 0.$$

The endpoint limits follow from the leading term $(-1)^n q^{n(n-1)/2} a^n$ and the zero at $a = 1$.

On I_k , the sum

$$S_{1,n}(a; q) = \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}$$

is strictly increasing because

$$\frac{\partial S_{1,n}}{\partial a} = S_{2,n} > 0.$$

As $a \downarrow q^{-k}$, the $j = k$ summand tends to $-\infty$, while as $a \uparrow q^{-(k+1)}$, the $j = k+1$ summand tends to $+\infty$. Thus there is exactly one zero of $S_{1,n}$, hence one critical point. At that point $S_{2,n} > 0$, so differentiating $P'_n = -P_n S_{1,n}$ shows $P''_n = -P_n S_{2,n} \neq 0$; the critical point is simple. Exactly $k+1$ factors are negative on I_k , giving the sign.

On the last interval, $S_{1,n}$ rises from $-\infty$ to 0^- , so it never vanishes. The sign and endpoint behavior give the final assertion. $\square$

Corollary 4.2 (PAPER PROOF Real root count at a regular target). *Let $v_{n,k} = P_n(c_{n,k})$. Every interior interval I_k contributes two real inverse values when the target lies strictly between 0 and $v_{n,k}$, one double inverse value when $y = v_{n,k}$, and none otherwise. The principal interval contributes one value for $y > 0$; the final interval contributes one value whenever $(-1)^n y > 0$. These contributions are independent and exhaust all real roots of $P_n(a) = y$.*

This is a complete real atlas. In particular, “the inverse finite q -Pochhammer symbol” is naturally a collection of branches indexed by the zero intervals.

4.2 Exact low-degree inverses

For $n = 1$, $(a; q)_1 = 1 - a$, so $a = 1 - y$. For $n = 2$,

$$(a; q)_2 = 1 - (1 + q)a + qa^2,$$

and therefore

$$a_{\pm}(y; q) = \frac{1 + q \pm \sqrt{(1 - q)^2 + 4qy}}{2q}. \quad (4.1)$$

The discriminant in y vanishes at $-(1 - q)^2/(4q)$, the unique critical value. Formula (4.1) is the model for every simple branch point: two sheets exchange when the target circles the critical value.

Whenever the hypotheses of the generic symmetric-monodromy proposition below hold, the monodromy is S_n ; for $n \geq 5$, that branch problem is not solvable by radicals. The uniform generic-Galois assertion remains conjectural. This is not a deficiency of the theory: discriminants, monodromy, convergent local series, and certified continuation provide the appropriate replacement.

4.3 The principal series at $y = 1$

Euler's finite q -binomial theorem gives

$$P_n(a) = \sum_{m=0}^n (-1)^m q^{\binom{m}{2}} \begin{bmatrix} n \\ m \end{bmatrix}_q a^m.$$

Set

$$e_1 = [n]_q, \quad e_2 = q \begin{bmatrix} n \\ 2 \end{bmatrix}_q, \quad e_3 = q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q, \quad u = 1 - y.$$

Proposition 4.3 (PAPER PROOF Local inverse at the origin). *The branch with $a(1) = 0$ has expansion*

$$a = \frac{u}{e_1} + \frac{e_2}{e_1^3} u^2 + \left(\frac{2e_2^2}{e_1^5} - \frac{e_3}{e_1^4} \right) u^3 + O(u^4). \quad (4.2)$$

All higher coefficients are obtained recursively from the finite q -binomial coefficients, or directly by Lagrange–Buerhmann inversion.

Proof. The direct series is $u = e_1 a - e_2 a^2 + e_3 a^3 + O(a^4)$. Substitute $a = A_1 u + A_2 u^2 + A_3 u^3 + O(u^4)$ and compare coefficients. $\square$

Because the nearest singularity of this branch is a critical value, the exact radius of convergence is the distance from 1 to the closest critical value that is visible on the analytic continuation of the sheet. Determining that visibility is a monodromy problem, not merely a nearest-point calculation.

4.4 The discriminant of the product

Theorem 4.4 (PAPER PROOF Exact zero-level discriminant). *For $n \geq 1$,*

$$\text{Disc}_a P_n(a; q) = q^{n(n-1)(n-2)/3} \prod_{d=1}^{n-1} (1 - q^d)^{2(n-d)}. \quad (4.3)$$

Proof. The leading coefficient is $L = (-1)^n q^N$, $N = n(n-1)/2$, and the roots are q^{-j} . Hence

$$\text{Disc } P_n = L^{2n-2} \prod_{0 \leq i < j < n} (q^{-i} - q^{-j})^2.$$

Now

$$q^{-i} - q^{-j} = -q^{-j}(1 - q^{j-i}).$$

For each difference $d = j - i$, the factor $(1 - q^d)^2$ occurs $n - d$ times. The total exponent of q is

$$n(n-1)^2 - 2 \sum_{j=1}^{n-1} j^2 = \frac{n(n-1)(n-2)}{3}.$$

□

For a variable target define

$$\Delta_n(y; q) = \text{Disc}_a(P_n(a; q) - y).$$

Proposition 4.5 (PAPER PROOF Critical-value factorization). *Let $c_1, \dots, c_{n-1}$ be the roots of P'_n , counted with multiplicity. Then*

$$\Delta_n(y; q) = (-1)^{n(n-1)/2} n^n L^{n-1} \prod_{j=1}^{n-1} (P_n(c_j; q) - y), \quad L = (-1)^n q^{n(n-1)/2}. \quad (4.4)$$

Proof. Use

$$\text{Disc}(F) = (-1)^{n(n-1)/2} L^{-1} \text{Res}(F, F')$$

and evaluate the resultant through the roots of $F' = P'_n$, whose leading coefficient is nL . □

Equation (4.4) is the algebraic bridge between forward critical-point calculations and inverse branch geometry.

4.5 Secondary discriminants and the first Maxwell collision

Define

$$\mathfrak{M}_n(q) = \text{Disc}_y \Delta_n(y; q).$$

For $0 < q < 1$, all critical points are simple by the real branch theorem. Therefore $\mathfrak{M}_n(q) = 0$ in this interval if and only if two distinct critical points have the same critical value.

Exact symbolic factorization gives

$$\mathfrak{M}_3(q) = 16q^6(q-1)^6(q^2+q+1)^3, \quad (4.5)$$

$$\begin{aligned} \mathfrak{M}_4(q) &= 16q^{42}(q-1)^{26}(q+1)^6(q^2+1)^8 \\ &\quad \times (9q^4 + 31q^3 + 45q^2 + 31q + 9)^3, \end{aligned} \quad (4.6)$$

and

$$\begin{aligned} \mathfrak{M}_5(q) &= 256q^{150}(q-1)^{60}(q+1)^{14}(q^2+q+1)^4(q^4+q^3+q^2+q+1)^{15} \\ &\quad \times (36q^6 + 97q^5 + 181q^4 + 213q^3 + 181q^2 + 97q + 36)^3 \\ &\quad \times H_5(q)^2, \end{aligned} \quad (4.7)$$

where

$$\begin{aligned} H_5(q) &= 8q^{10} - 5q^9 - 7q^8 - 6q^7 - 8q^6 + 35q^5 \\ &\quad - 8q^4 - 6q^3 - 7q^2 - 5q + 8. \end{aligned} \quad (4.8)$$

Theorem 4.6 (PAPER PROOF First positive-base Maxwell parameter). *The polynomial H_5 has exactly one zero q_0 in $(0, 1)$. At this parameter the polynomial map $a \mapsto (a; q_0)_5$ has two distinct simple critical points with a common critical value. No such collision occurs for $n = 3$ or $n = 4$ and $0 < q < 1$.*

Proof. The factors in (4.5) and (4.6) are nonzero on $(0, 1)$, so the first assertion for $n = 3, 4$ follows. The polynomial H_5 is reciprocal. Put $x = q + q^{-1}$. Dividing by q^5 gives

$$\frac{H_5(q)}{q^5} = h(x) = 8x^5 - 5x^4 - 47x^3 + 14x^2 + 53x + 37.$$

For $0 < q < 1$, one has $x > 2$. Now $h(2) = -1$, while $h(x) \rightarrow +\infty$. Moreover,

$$h'(x) = 40x^4 - 20x^3 - 141x^2 + 28x + 53,$$

$$h''(x) = 160x^3 - 60x^2 - 282x + 28, \quad h'''(x) = 480x^2 - 120x - 282.$$

At $x = 2$, all three are positive, and $h^{(4)}(x) = 960x - 120 > 0$ for $x \geq 2$. It follows successively that h''' , h'' , h' , and then h are strictly increasing on $[2, \infty)$. Thus h has exactly one root there, which corresponds to one reciprocal pair of positive roots of H_5 , exactly one lying in $(0, 1)$.

At this root all the other factors in (4.7) are nonzero. Since the critical points are simple for positive $q < 1$, vanishing of the secondary discriminant means that two distinct critical values coincide. $\square$

Computational result 4.7 (SYMBOLIC/NUMERICAL Identification of the colliding sheets). High-precision computation gives

$$\begin{aligned} q_0 &= 0.84015337830837665866293643188913294310139258 \dots, \\ c_{5,0}(q_0) &= 1.0714044413496100275076056970032887643 \dots, \\ c_{5,2}(q_0) &= 1.5701685824623346994394512430684932918 \dots, \end{aligned}$$

and

$$P_5(c_{5,0}; q_0) = P_5(c_{5,2}; q_0) = -0.0002954183213260369789427474653292421974 \dots$$

The residual in the archived 60-digit calculation is below 4×10^{-50} . The exact factorization proves the existence and uniqueness of the positive collision parameter; the sheet identification is reproducible numerical evidence and can be upgraded to a formal interval proof.

For $n = 6$, the exact secondary discriminant has another reciprocal factor with a root

$$q \approx 0.77737031523884893092862981385481448,$$

again corresponding numerically to the collision of the I_0 and I_2 critical values. The full factor is recorded in `data/secondary_discriminants.txt`.

4.6 Monodromy away from the Maxwell set

Proposition 4.8 (PAPER PROOF Generic symmetric monodromy). *Fix n and a base q for which all critical points of P_n are simple and all critical values are distinct. Then the monodromy group of $P_n(a) = y$ over the regular y -plane is the full symmetric group S_n . In particular this holds for every $q \in (0, 1)$ for which $\mathfrak{M}_n(q) \neq 0$.*

Proof. A small loop around a simple critical value exchanges exactly the two sheets meeting there, hence gives a transposition. The polynomial covering is connected, so the monodromy action is transitive. A transitive permutation group generated by transpositions is the full symmetric group: form the graph whose vertices are sheets and whose edges are the generating transpositions; transitivity makes the graph connected, and edge transpositions of a connected graph generate S_n . $\square$

At a Maxwell parameter, a loop around the common critical value can produce a product of two disjoint transpositions. Splitting the parameter away from the Maxwell locus resolves it into two ordinary branch values. This is the appropriate local unfolding.

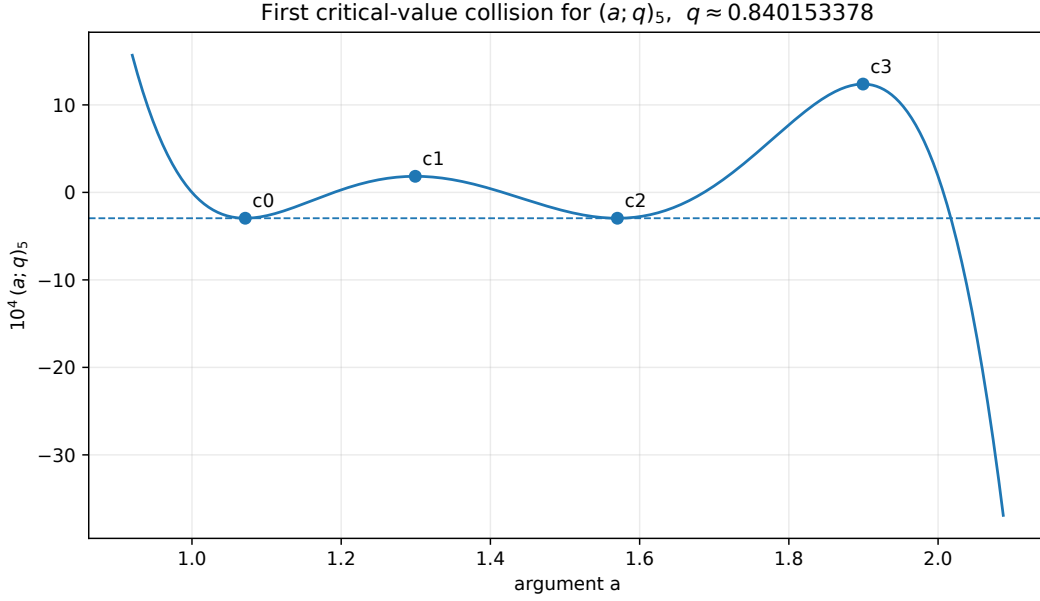


Figure 4.1: The first positive-base collision of distinct critical values for a finite q -Pochhammer product. At $q = q_0$, the extrema in the intervals I_0 and I_2 lie on the same horizontal level.

5 The infinite product: inversion in the argument

5.1 The principal inverse

For $0 < q < 1$, the entire function

$$P(a) = (a; q)_\infty$$

has simple zeros q^{-j} , $j \geq 0$. On the zero-free half-line $a < 1$, it has a canonical inverse.

Definition 5.1 (Principal inverse product). For $0 < q < 1$, define $\mathcal{A}_q : (0, \infty) \rightarrow (-\infty, 1)$ by

$$(\mathcal{A}_q(y); q)_\infty = y.$$

Theorem 5.2 (PAPER PROOF Global principal branch). *The map $a \mapsto (a; q)_\infty$ is a real-analytic, strictly decreasing bijection from $(-\infty, 1)$ onto $(0, \infty)$. Consequently $\mathcal{A}_q$ is real analytic, strictly decreasing, and convex. If*

$$S_r(a) = \sum_{j \geq 0} \frac{q^{rj}}{(1 - aq^j)^r},$$

then

$$\mathcal{A}'_q(y) = -\frac{1}{yS_1(\mathcal{A}_q(y))}, \quad (5.1)$$

$$\mathcal{A}''_q(y) = \frac{S_1(\mathcal{A}_q(y))^2 - S_2(\mathcal{A}_q(y))}{y^2 S_1(\mathcal{A}_q(y))^3} > 0. \quad (5.2)$$

Proof. Every factor is positive for $a < 1$, and the logarithmic derivative is $-S_1 < 0$. The product tends to zero as $a \uparrow 1$. As $a \rightarrow -\infty$, every finite truncation tends to infinity and is bounded above by the infinite product, so the latter tends to infinity. The inverse exists. Formula (5.1) follows from implicit differentiation. Since

$$S_1^2 - S_2 = 2 \sum_{0 \leq i < j} \frac{q^{i+j}}{(1 - aq^i)(1 - aq^j)} > 0,$$

formula (5.2) proves convexity. □

5.2 Three distinguished target regimes

5.2.1 Near $y = 1$

The infinite q -binomial theorem gives

$$(a; q)_\infty = 1 - \frac{a}{1-q} + \frac{qa^2}{(1-q)(1-q^2)} - \frac{q^3a^3}{(1-q)(1-q^2)(1-q^3)} + \cdots.$$

Proposition 5.3 (PAPER PROOF Central principal series). *With $u = 1 - y$,*

$$\begin{aligned} \mathcal{A}_q(y) &= (1-q)u + \frac{q(1-q)}{1+q}u^2 \\ &\quad + \frac{q^2(1-q)(q^2+q+2)}{(1+q)^2(q^2+q+1)}u^3 + O(u^4). \end{aligned} \quad (5.3)$$

Proof. Apply the three-term series-reversion formula to the first three nonconstant coefficients of the direct product. □

5.2.2 Near the first zero, $y = 0^+$

Let

$$R_q = (q; q)_\infty, \quad L_q = \sum_{j=1}^{\infty} \frac{q^j}{1-q^j}.$$

Writing $a = 1 - \delta$ gives

$$(a; q)_\infty = \delta(aq; q)_\infty = R_q \delta(1 + L_q \delta + O(\delta^2)).$$

Hence

$$\mathcal{A}_q(y) = 1 - \frac{y}{R_q} + L_q \left(\frac{y}{R_q} \right)^2 + O(y^3). \quad (5.4)$$

This expansion is especially useful when solving Euler q -exponential equations at large positive values.

5.2.3 Large targets and log-periodic correction

The behavior as $y \rightarrow \infty$ is not a simple power law. Put $q = e^{-\lambda}$, $\lambda > 0$, and write $x = -a > 0$, $u = \log x$. Choose the unique decomposition

$$x = q^{-m}r, \quad m \in \mathbb{N}_0, \quad 1 \leq r < q^{-1},$$

and set $\rho = \log r \in [0, \lambda)$.

Theorem 5.4 (PAPER PROOF Theta-modulated large-target law). *Define the bounded λ -periodic phase function*

$$\Omega_q(\rho) = \log((-r; q)_\infty(-q/r; q)_\infty) - \frac{\rho}{2} - \frac{\rho^2}{2\lambda}, \quad r = e^\rho.$$

Then, as $x \rightarrow \infty$,

$$\log(-x; q)_\infty = \frac{u^2}{2\lambda} + \frac{u}{2} + \Omega_q(u \bmod \lambda) + O(e^{-u}). \quad (5.5)$$

Consequently, as $y \rightarrow \infty$,

$$\log(-\mathcal{A}_q(y)) = \sqrt{2\lambda \log y} - \frac{\lambda}{2} + O((\log y)^{-1/2}), \quad (5.6)$$

with a computable log-periodic correction at the displayed error scale.

Proof. The exact finite shift is

$$\begin{aligned} (-q^{-m}r; q)_\infty &= \prod_{j=0}^{m-1} (1 + rq^{j-m})(-r; q)_\infty \\ &= r^m q^{-m(m+1)/2} (-q/r; q)_m (-r; q)_\infty. \end{aligned}$$

The finite product $(-q/r; q)_m$ differs from its infinite limit by a relative factor $1 + O(q^m)$, uniformly for $1 \leq r \leq q^{-1}$. Taking logarithms and using $u = m\lambda + \rho$ yields

$$\frac{\lambda m(m+1)}{2} + m\rho = \frac{u^2}{2\lambda} + \frac{u}{2} - \frac{\rho}{2} - \frac{\rho^2}{2\lambda}.$$

This proves (5.5). Complete the square in the first two terms and invert. Since Ω_q is bounded, it perturbs u by $O((\log y)^{-1/2})$. $\square$

The periodic phase is a genuine lattice effect. Replacing the product sum by a single Euler-Maclaurin integral captures the quadratic envelope but misses this fine structure.

5.3 All nonprincipal real branches

For every $k \geq 0$, let

$$I_k = (q^{-k}, q^{-(k+1)}).$$

The same proof as in the finite case gives one simple critical point $c_k(q) \in I_k$. The sign on I_k is $(-1)^{k+1}$, and both endpoints map to zero. Thus a target strictly between zero and the signed extremum has two inverse values in that interval.

Theorem 5.5 (PAPER PROOF Asymptotics of remote branch points). *Let $m = k + 1$. As $k \rightarrow \infty$,*

$$c_k(q) = q^{-m} \left(1 - \frac{1}{m+1} + O(m^{-2}) \right), \quad (5.7)$$

and

$$\begin{aligned} |(c_k(q); q)_\infty| &= (q; q)_\infty^2 q^{-m(m+1)/2} \frac{m^m}{(m+1)^{m+1}} (1 + O(m^{-1})) \\ &= \frac{(q; q)_\infty^2}{e(m+1)} q^{-m(m+1)/2} (1 + O(m^{-1})). \end{aligned} \quad (5.8)$$

The constants implied by the O -terms may be chosen locally uniformly for q in compact subsets of $(0, 1)$.

Proof. Write

$$a = q^{-m}(1 - \varepsilon), \quad 0 < \varepsilon < 1 - q.$$

A direct split at the zero index gives the exact identity

$$\begin{aligned} (a; q)_\infty &= (-1)^m q^{-m(m+1)/2} \varepsilon (1 - \varepsilon)^m G_m(\varepsilon), \\ G_m(\varepsilon) &= \left(\frac{q}{1 - \varepsilon}; q \right)_m ((1 - \varepsilon)q; q)_\infty. \end{aligned} \quad (5.9)$$

For $\varepsilon = O(m^{-1})$, the logarithmic derivative of G_m is uniformly bounded and

$$G_m(\varepsilon) = (q; q)_\infty^2 (1 + O(m^{-1})).$$

At the extremum,

$$0 = \frac{1}{\varepsilon} - \frac{m}{1 - \varepsilon} + \frac{\partial}{\partial \varepsilon} \log G_m(\varepsilon).$$

The bounded final term perturbs the solution of the first two terms by $O(m^{-2})$, so $\varepsilon = 1/(m+1) + O(m^{-2})$. Substitution in (5.9) yields (5.8); Stirling's elementary limit $(1 - 1/(m+1))^m = e^{-1}(1 + O(m^{-1}))$ gives the second line. $\square$

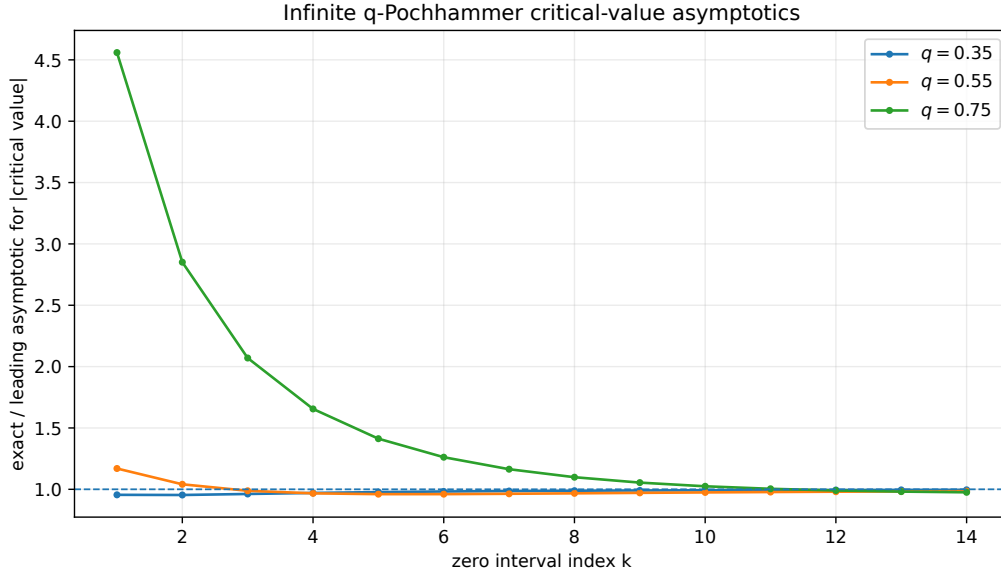


Figure 5.1: Ratio of exact nonprincipal critical-value magnitudes to the leading asymptotic in (5.8). Convergence is slower when q is close to one, as expected from the product tail.

At each $v_k = (c_k; q)_\infty$, the two local branches obey the universal square-root law

$$a = c_k \pm \sqrt{\frac{2(y - v_k)}{P''(c_k)}} + O(y - v_k).$$

Equations (5.7)–(5.8) therefore give an asymptotic atlas of the entire real inverse surface.

5.4 A Bernstein-function conjecture

Let

$$X_q(y) = 1 - \mathcal{A}_q(y), \quad y > 0.$$

Then $X_q > 0$, $X'_q > 0$, and $X''_q < 0$. The archived series-reversion experiment finds

$$(-1)^{r-1} X_q^{(r)}(y) > 0, \quad 1 \leq r \leq 8,$$

at fifteen representative points with $q \in \{0.25, 0.5, 0.75\}$ and arguments ranging from -3 to 0.8 .

Conjecture 5.6 (**CONJECTURAL** Complete Bernstein principal inverse). *For every $0 < q < 1$, the function $X_q = 1 - \mathcal{A}_q$ is a complete Bernstein function on $(0, \infty)$. Equivalently, X'_q is completely monotone and X_q admits a Pick–Nevanlinna continuation with the standard complete Bernstein representation.*

A proof would simultaneously control all inverse derivatives, provide sharp continued-fraction approximants, and connect the zero sequence q^{-j} to Stieltjes moment theory. The conjecture is substantially stronger than the proved convexity of $\mathcal{A}_q$.

6 Inversion with respect to the base q

6.1 Finite products on the principal sign domains

Fix $a \in \mathbb{R}$ and $n \geq 2$. Regard $(a; q)_n$ as a polynomial in q of degree $n(n-1)/2$.

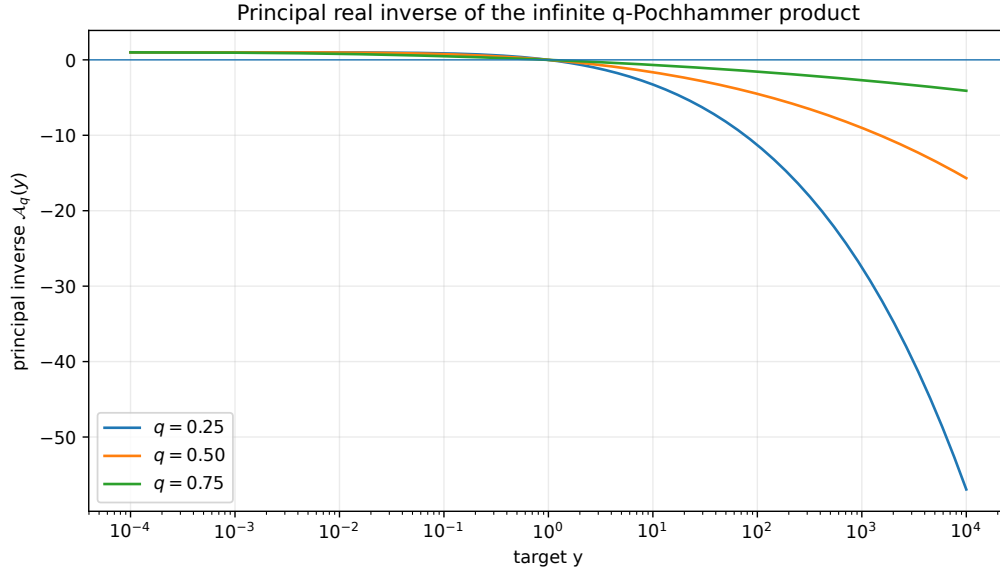


Figure 5.2: The principal inverse $\mathcal{A}_q(y)$. It decreases from 1 to $-\infty$, crossing zero at $y = 1$.

Theorem 6.1 (PAPER PROOF Global finite-base inverses). (i) If $0 < a < 1$, the map $q \mapsto (a; q)_n$ is strictly decreasing on $(0, 1)$, from $1 - a$ to $(1 - a)^n$. Thus there is exactly one real inverse $q \in (0, 1)$ for

$$(1 - a)^n < y < 1 - a.$$

(ii) If $a < 0$, the same map is strictly increasing on $(0, 1)$, from $1 - a$ to $(1 - a)^n$. Thus there is exactly one inverse for

$$1 - a < y < (1 - a)^n.$$

(iii) If $a > 1$, the real zeros $q = a^{-1/j}$, $1 \leq j \leq n - 1$, divide $(0, 1)$ into sign intervals. On each interval inverse branches are governed by the zeros of the explicit base derivative kernel; no global monotonicity claim is valid without specifying the interval.

Proof. For $0 < a < 1$, every factor with $j \geq 1$ decreases strictly with q , while the $j = 0$ factor is fixed and positive. For $a < 0$, every factor increases. The endpoint limits are immediate. For $a > 1$, factors vanish at the stated bases and the logarithmic derivative has poles there. $\square$

6.2 The small-base inverse series

Normalize

$$s = \frac{1 - y/(1 - a)}{a}, \quad a \neq 0.$$

For a target near $1 - a$, one obtains

$$s = q + q^2 - aq^3 + O(q^4), \quad n = 3,$$

and

$$s = q + q^2 + (1 - a)q^3 + O(q^4), \quad n \geq 4.$$

Therefore

$$q = s - s^2 + (2 + a)s^3 + O(s^4), \quad n = 3, \quad (6.1)$$

$$q = s - s^2 + (1 + a)s^3 + O(s^4), \quad n \geq 4. \quad (6.2)$$

For $n = 2$, the exact answer is $q = s$. These formulas also apply to the infinite product because factors beginning with q^4 do not affect the cubic truncation.

6.3 The infinite product as a function of the base

Theorem 6.2 (PAPER PROOF Global infinite-base inverse). *Fix $a \in \mathbb{R}$.*

- (i) *If $0 < a < 1$, then $q \mapsto (a; q)_\infty$ is strictly decreasing on $(0, 1)$, from $1 - a$ to 0. Every $y \in (0, 1 - a)$ has exactly one base inverse, denoted $Q_a(y)$.*
- (ii) *If $a < 0$, the map is strictly increasing on $(0, 1)$, from $1 - a$ to $+\infty$. Every $y > 1 - a$ has exactly one base inverse.*
- (iii) *If $a = 0$, the product is identically one.*

Proof. The derivative kernel has the asserted sign. For $0 < a < 1$, as $q \rightarrow 1^-$, infinitely many factors are bounded above by a fixed number less than one, so the product tends to zero. For $a < 0$, arbitrarily many factors exceed $1 + |a|/2$ when q is sufficiently close to one, so the product tends to infinity. The $q \rightarrow 0$ limit is $1 - a$. $\square$

6.4 Polylogarithmic inversion near $q = 1$

Let $0 < a < 1$, put $q = e^{-t}$, and define

$$A = \text{Li}_2(a), \quad B = \frac{1}{2} \log(1 - a), \quad C = \frac{a}{12(1 - a)}.$$

The classical expansion of the shifted factorial is

$$\log(a; e^{-t})_\infty = -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) - \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} \text{Li}_{2-2r}(a) t^{2r-1}. \quad (6.3)$$

It is asymptotic in sectors and, after truncation, admits standard remainder bounds; see McIntosh [5] and later Lambert-series treatments [7].

Theorem 6.3 (PAPER PROOF Base inverse as $y \rightarrow 0^+$). *Let*

$$X = B - \log y = \log \frac{\sqrt{1 - a}}{y}.$$

Then the unique base inverse satisfies

$$t = -\log Q_a(y) = \frac{A}{X} + \frac{CA^2}{X^3} + O(X^{-5}), \quad y \rightarrow 0^+, \quad (6.4)$$

so $Q_a(y) = \exp(-t)$. Higher terms follow by formal reversion of (6.3).

Proof. From (6.3),

$$X = \frac{A}{t} + Ct + O(t^3).$$

Insert $t = A/X + d/X^3 + O(X^{-5})$. Expanding $A/t + Ct$ gives

$$X + \frac{-d/A + CA}{X} + O(X^{-3}),$$

so $d = CA^2$. $\square$

The dominant inverse scale is rational in $1/\log(1/y)$, not algebraic in y . Joint limits $a \rightarrow 1$ and $q \rightarrow 1$ can change this scale and create Lambert- W transition formulas; they are listed among the frontier problems in Section 18.

6.5 Complex base branches

For finite n , solving for q is algebraic of degree $n(n-1)/2$, with branch points given by

$$\text{Disc}_q((a; q)_n - y) = 0.$$

The reciprocal identity

$$(a; q)_n = (-a)^n q^{n(n-1)/2} (a^{-1}; q^{-1})_n \quad (6.5)$$

transports large-base behavior to small reciprocal base, provided zeros and logarithm branches are tracked.

For the infinite product, $q \mapsto (a; q)_\infty$ is analytic in $|q| < 1$. Its inverse branches can be continued until either $\partial_q (a; q)_\infty = 0$ or the path meets a singular boundary. Roots of unity organize the boundary asymptotics, but a blanket assertion of a natural boundary depends on the chosen parameter and normalization and is not made here.

7 Inversion with respect to order

7.1 Exact reduction to the principal argument inverse

Fix $0 < q < 1$, and initially let $0 < a < 1$. Put

$$P(a) = (a; q)_\infty, \quad (a; q)_\alpha = \frac{P(a)}{P(aq^\alpha)}.$$

Let

$$\alpha_0 = \log_q(1/a) < 0.$$

Theorem 7.1 (PAPER PROOF Real order inverse). *On (α_0, ∞) , the function $\alpha \mapsto (a; q)_\alpha$ is strictly decreasing and log-convex, from $+\infty$ to $P(a)$. For every $y > P(a)$, its unique real inverse is*

$$\alpha = \frac{\log(\mathcal{A}_q(P(a)/y)/a)}{\log q}. \quad (7.1)$$

On the nonnegative-order subbranch $\alpha \geq 0$, the target range is $P(a) < y \leq 1$; the lower endpoint is approached only as $\alpha \rightarrow \infty$.

Proof. The derivative formulas in the previous chapter show strict decrease and log-convexity while $0 < aq^\alpha < 1$. At $\alpha \downarrow \alpha_0$, the denominator has its first zero and the quotient diverges. At infinity the denominator tends to one. Solving

$$P(aq^\alpha) = \frac{P(a)}{y}$$

gives $aq^\alpha = \mathcal{A}_q(P(a)/y)$, hence (7.1). $\square$

For $a < 0$, the same algebraic reduction remains valid on the real branch, with monotonicity reversed: $(a; q)_\alpha$ increases from 1 at $\alpha = 0$ toward the unattained limit $(a; q)_\infty > 1$ as $\alpha \rightarrow \infty$.

7.2 Complex order lattice

Because $q^\alpha = \exp(\alpha \log q)$, the complex order is periodic under

$$\alpha \mapsto \alpha + \frac{2\pi im}{\log q}, \quad m \in \mathbb{Z}.$$

If b is any chosen inverse value satisfying $(b; q)_\infty = P(a)/y$, then the order sheets are

$$\alpha_m = \frac{\text{Log}(b/a) + 2\pi im}{\log q}, \quad m \in \mathbb{Z}. \quad (7.2)$$

Thus order inversion factors into two independent branch structures: the infinite-product inverse selecting b , and the logarithmic lattice selecting m .

7.3 Discrete order recovery

For $0 < a < 1$, the sequence $(a; q)_n$ decreases strictly from 1 to $(a; q)_\infty$. A target has an exact integer inverse only if it belongs to this sequence. For arbitrary targets in the interval, the continuous order inverse (7.1) supplies a canonical interpolation; its floor and ceiling bracket the target through

$$(a; q)_{\lceil \alpha \rceil} \leq y \leq (a; q)_{\lfloor \alpha \rfloor}.$$

This is preferable to an ad hoc nearest-integer search because it preserves sensitivity information and extends analytically.

7.4 Multiple shifted factors

For

$$F(\alpha) = \prod_{r=1}^s (a_r; q)_\alpha^{\mu_r},$$

where the exponents μ_r are fixed, the logarithmic order derivative is

$$\frac{F'(\alpha)}{F(\alpha)} = \log q \sum_{r=1}^s \mu_r \sum_{j \geq 0} \frac{a_r q^{\alpha+j}}{1 - a_r q^{\alpha+j}}.$$

Any sign theorem for the combined Lambert kernel yields a global inverse; in the absence of such a sign, the zeros of this kernel divide the order line into monotone branches. This single formula covers many ratios occurring in basic hypergeometric coefficients.

Part III

Gaussian coefficients and combinatorial q -analogues

8 Inverting a Gaussian coefficient with respect to q

8.1 A global positive-base theorem

For integers $0 \leq k \leq n$, write

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^D c_j q^j, \quad D = k(n-k).$$

The coefficients c_j are nonnegative integers, $c_0 = c_D = 1$, and $c_j = c_{D-j}$.

Theorem 8.1 (PAPER PROOF Unique positive-base inverse). *Let $0 < k < n$. The map $q \mapsto G_{n,k}(q)$ is strictly increasing on $(0, \infty)$, from 1 to $+\infty$. Hence for every target $y > 1$ there is exactly one positive real solution, denoted*

$$q = \mathcal{Q}_{n,k}(y).$$

Moreover,

$$\begin{cases} 0 < \mathcal{Q}_{n,k}(y) < 1, & 1 < y < \binom{n}{k}, \\ \mathcal{Q}_{n,k}(y) = 1, & y = \binom{n}{k}, \\ \mathcal{Q}_{n,k}(y) > 1, & y > \binom{n}{k}. \end{cases}$$

Proof. Since $D > 0$, at least one coefficient with positive exponent is nonzero, and

$$G'_{n,k}(q) = \sum_{j=1}^D j c_j q^{j-1} > 0$$

for $q > 0$. The endpoint values are $G(0) = 1$, $G(1) = \binom{n}{k}$, and $G(q) \sim q^D$ as $q \rightarrow \infty$. $\square$

This theorem is stronger than a local implicit-function result: it gives a single globally defined positive branch for every nontrivial Gaussian coefficient and every admissible target.

8.2 Expansion at $q = 0$

Let

$$G_{n,k}(q) = 1 + q + c_2 q^2 + c_3 q^3 + \cdots,$$

where c_j counts partitions of j fitting in a $k \times (n-k)$ rectangle. For $u = y - 1$, series reversion gives

$$\mathcal{Q}_{n,k}(y) = u - c_2 u^2 + (2c_2^2 - c_3) u^3 + O(u^4). \quad (8.1)$$

Here c_2 is the number of the partitions (2) and (1, 1) that fit the rectangle. More generally, every inverse coefficient is an integer polynomial in the rectangle-partition numbers. Determining its sign pattern is an interesting combinatorial reversion problem.

8.3 Complete Bernoulli reversion near $q = 1$

Put $q = e^{-t}$, $D = k(n-k)$, and

$$B = \binom{n}{k}, \quad s = -\log \frac{y}{B}.$$

For the $q < 1$ branch, $s > 0$. The forward expansion is

$$\log \frac{G_{n,k}(e^{-t})}{B} = -\frac{D}{2}t + \sum_{r \geq 1} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^k ((n-k+j)^{2r} - j^{2r}). \quad (8.2)$$

This is the Bernoulli expansion established in the source monograph and is convergent for fixed n, k when $|t| < 2\pi/n$.

Theorem 8.2 (PAPER PROOF First inverse terms at $q = 1$). As $y \uparrow B$ from below,

$$t = -\log \mathcal{Q}_{n,k}(y) = \frac{2s}{D} + \frac{n+1}{3D^2} s^2 + \frac{(n+1)^2}{9D^3} s^3 + O(s^4). \quad (8.3)$$

For the branch $q > 1$, put $r = \log(y/B) > 0$; then

$$\log \mathcal{Q}_{n,k}(y) = \frac{2r}{D} - \frac{n+1}{3D^2} r^2 + \frac{(n+1)^2}{9D^3} r^3 + O(r^4). \quad (8.4)$$

Proof. The coefficient of t^2 in (8.2) is

$$\frac{1}{24} \sum_{j=1}^k ((n-k+j)^2 - j^2) = \frac{D(n+1)}{24}.$$

Thus

$$s = \frac{D}{2}t - \frac{D(n+1)}{24}t^2 + O(t^4).$$

There is no cubic term because the coefficient distribution is symmetric about $D/2$. Reversion gives (8.3). Replacing t by $-t$, or using palindromicity directly, gives (8.4). $\square$

The fourth inverse coefficient can also be written explicitly. Define

$$K_4 = \frac{1}{2880} \sum_{j=1}^k ((n-k+j)^4 - j^4).$$

If $A = D/2$ and $C = D(n+1)/24$, then the coefficient of s^4 in $t(s)$ is

$$\frac{5C^3}{A^7} - \frac{K_4}{A^5}.$$

Higher coefficients follow from ordinary Bell-polynomial series reversion. Thus the forward Bernoulli expansion automatically generates a complete inverse expansion.

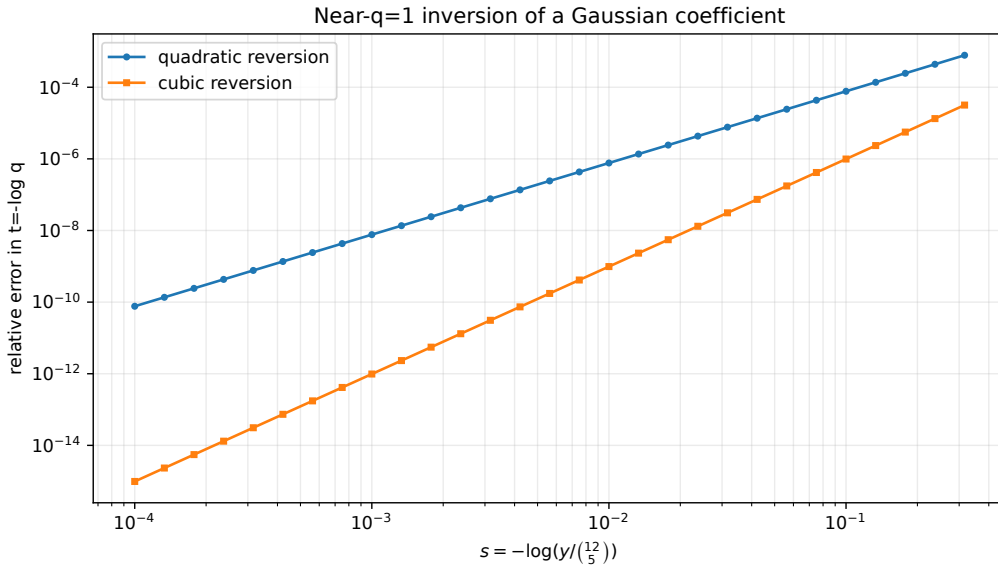


Figure 8.1: Relative errors of the quadratic and cubic truncations of (8.3) for $\begin{bmatrix} 12 \\ 5 \end{bmatrix}_q$. The observed slopes agree with the predicted next powers.

8.4 The large-target inverse

Palindromicity gives

$$G_{n,k}(q) = q^D G_{n,k}(q^{-1}). \quad (8.5)$$

Since the coefficient of q is one,

$$G_{n,k}(q) = q^D (1 + q^{-1} + O(q^{-2})).$$

Proposition 8.3 (PAPER PROOF Large-target expansion). As $y \rightarrow \infty$,

$$\mathcal{Q}_{n,k}(y) = y^{1/D} - \frac{1}{D} + O(y^{-1/D}). \quad (8.6)$$

A full expansion in powers of $y^{-1/D}$ is obtained by reverting (8.5) at $q^{-1} = 0$.

Proof. Put $Y = y^{1/D}$ and $q = Y + c + O(Y^{-1})$. Then $q^D = Y^D(1 + Dc/Y + O(Y^{-2}))$ and $1 + q^{-1} = 1 + Y^{-1} + O(Y^{-2})$. Equality with $y = Y^D$ forces $Dc + 1 = 0$. $\square$

8.5 Complex branches and cyclotomic geometry

The positive real branch is only one sheet of the degree- D algebraic inverse. The cyclotomic factorization

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d \geq 2} \Phi_d(q)^{\lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor} \quad (8.7)$$

shows how roots of unity enter the zero set. Critical values are the roots of

$$\text{Disc}_q(G_{n,k}(q) - y).$$

Analytic continuation of $\mathcal{Q}_{n,k}$ around those values permutes the D complex sheets. Reciprocal symmetry relates the covering near zero to the covering near infinity but does not identify equal-target sheets directly, because the target is rescaled by q^D .

9 Inverting the upper and lower Gaussian parameters

9.1 Discrete parameters n and k

Fix $q > 0$. The standard ratio identities give

$$\frac{\begin{bmatrix} n+1 \\ k \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{[n+1]_q}{[n+1-k]_q} > 1, \quad 0 < k \leq n, \quad (9.1)$$

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{[n-k]_q}{[k+1]_q}. \quad (9.2)$$

Because $[x]_q$ is increasing in a positive integer x , the sequence in n is strictly increasing for fixed k , while the sequence in k is symmetric and strictly increases up to the middle, then decreases.

Theorem 9.1 (PAPER PROOF Discrete inverse structure). *For fixed $q > 0$ and $k \geq 1$, $n \mapsto \begin{bmatrix} n \\ k \end{bmatrix}_q$ is strictly increasing for $n \geq k$. For $0 < q < 1$, it converges to $1/(q; q)_k$; for $q \geq 1$ it is unbounded (at $q = 1$, this is the usual classical binomial growth).*

For fixed n , the map $k \mapsto \begin{bmatrix} n \\ k \end{bmatrix}_q$ is symmetric under $k \leftrightarrow n - k$, strictly increasing for $k < (n - 1)/2$, and strictly decreasing after the center. A target therefore has at most two integer k -preimages, paired by symmetry.

Proof. The monotonicity follows from (9.1)–(9.2). For $0 < q < 1$, use the product formula and let $n \rightarrow \infty$: each numerator factor tends to one, leaving $1/(q; q)_k$. $\square$

This discrete structure is exactly mirrored by the continuous interpolation below.

9.2 A two-parameter analytic interpolation

For $\alpha \geq \beta \geq 0$, define

$$\mathcal{G}_q(\alpha, \beta) = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}. \quad (9.3)$$

It agrees with $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at nonnegative integers.

9.2.1 Recovering the upper parameter

Theorem 9.2 (PAPER PROOF Upper-parameter inverse). *Fix $0 < q < 1$ and $\beta > 0$. The function $\alpha \mapsto \mathcal{G}_q(\alpha, \beta)$ is strictly increasing and strictly log-concave on $[\beta, \infty)$, from 1 to*

$$L_{q,\beta} = \frac{1}{(q; q)_\beta} = \frac{1}{(1-q)^\beta \Gamma_q(\beta+1)}. \quad (9.4)$$

Hence every $y \in (1, L_{q,\beta})$ has exactly one upper-parameter inverse.

Proof. Let $\psi_q = (\log \Gamma_q)'$. Then

$$\partial_\alpha \log \mathcal{G}_q = \psi_q(\alpha+1) - \psi_q(\alpha-\beta+1) > 0$$

because ψ_q is strictly increasing. Its derivative is

$$\psi'_q(\alpha+1) - \psi'_q(\alpha-\beta+1) < 0$$

because ψ'_q is strictly decreasing. The endpoint at $\alpha = \beta$ is one. The product representation

$$\mathcal{G}_q(\alpha, \beta) = \frac{(q^{\beta+1}; q)_\infty}{(q; q)_\infty} \frac{(q^{\alpha-\beta+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty}$$

gives the limit. □

Put $z = q^{\alpha-\beta+1}$. The exact factorization

$$\mathcal{G}_q(\alpha, \beta) = L_{q,\beta}(z; q)_\beta$$

reduces the upper-parameter inverse to order or argument inversion. As $y \uparrow L_{q,\beta}$,

$$L_{q,\beta} - y = L_{q,\beta}[\beta]_q z + O(z^2),$$

so

$$\alpha = \beta - 1 + \frac{\log((L_{q,\beta} - y)/(L_{q,\beta}[\beta]_q))}{\log q} + o(1). \quad (9.5)$$

The inverse diverges only logarithmically as the finite ceiling is approached.

9.2.2 Recovering the lower parameter

Theorem 9.3 (PAPER PROOF Two lower-parameter branches). *Fix $\alpha > 0$. The function $\beta \mapsto \mathcal{G}_q(\alpha, \beta)$ is symmetric about $\alpha/2$, strictly log-concave on $(0, \alpha)$, equals one at both endpoints, and has its unique maximum*

$$M_{q,\alpha} = \frac{\Gamma_q(\alpha+1)}{\Gamma_q(\alpha/2+1)^2}$$

at $\beta = \alpha/2$. Every $y \in (1, M_{q,\alpha})$ therefore has exactly two inverse branches $\beta_- < \alpha/2 < \beta_+$, related by $\beta_+ = \alpha - \beta_-$.

Proof. Symmetry is immediate. Differentiation gives

$$\partial_\beta \log \mathcal{G}_q = -\psi_q(\beta+1) + \psi_q(\alpha-\beta+1),$$

which vanishes only at the center, and

$$\partial_\beta^2 \log \mathcal{G}_q = -\psi'_q(\beta+1) - \psi'_q(\alpha-\beta+1) < 0.$$

□

Near the maximum, with $x_0 = \alpha/2 + 1$,

$$\beta_\pm(y) = \frac{\alpha}{2} \pm \sqrt{\frac{M_{q,\alpha} - y}{M_{q,\alpha} \psi'_q(x_0)}} + O(M_{q,\alpha} - y). \quad (9.6)$$

This is another explicit occurrence of the universal square-root inverse law.

9.3 Inverting the base in the generalized coefficient

For nonintegral α, β , positivity of polynomial coefficients is no longer available. The local derivative is

$$\partial_q \log \mathcal{G}_q = \partial_q \log \Gamma_q(\alpha + 1) - \partial_q \log \Gamma_q(\beta + 1) - \partial_q \log \Gamma_q(\alpha - \beta + 1).$$

At $q \rightarrow 1^-$, the first-order q -gamma expansion gives

$$\log \frac{\mathcal{G}_{e^{-t}}(\alpha, \beta)}{\binom{\alpha}{\beta}} = -\frac{\beta(\alpha - \beta)}{2}t + O(t^2), \quad (9.7)$$

where the classical generalized binomial coefficient is interpreted by gamma functions. Thus the same effective degree $D = \beta(\alpha - \beta)$ controls the leading inverse near $q = 1$. A global monotonicity theorem for all real $\alpha \geq \beta \geq 0$ would be valuable; the integer theorem proves it on the lattice but does not automatically interpolate.

Research problem 9.4. Determine the maximal region in the (α, β) -plane on which $q \mapsto \mathcal{G}_q(\alpha, \beta)$ is strictly increasing for $q > 0$, or strictly increasing on $(0, 1)$. Express any boundary in terms of a completely monotone Lambert kernel.

10 A positive-coefficient inversion principle

10.1 The general theorem

Theorem 10.1 (PAPER PROOF Positive-polynomial inverse principle). *Let*

$$F(q) = c_0 + c_1q + \cdots + c_Dq^D, \quad c_j \geq 0,$$

with at least one $c_j > 0$ for $j \geq 1$. Then F is strictly increasing on $(0, \infty)$, from c_0 to $+\infty$. Hence it has a unique positive inverse for every $y > c_0$. The inverse is analytic except at images of complex critical points and has condition number

$$\kappa(q) = \left| \frac{F(q)}{qF'(q)} \right|.$$

Proof. For $q > 0$, $F'(q) = \sum_{j \geq 1} jc_jq^{j-1} > 0$. The remaining claims follow from the inverse function theorem. $\square$

This elementary result is a powerful inverse theorem because many combinatorial q -analogs are polynomials with nonnegative coefficients.

10.2 Examples

10.2.1 q -factorials

For $n \geq 2$,

$$[n]_q! = \prod_{j=1}^n (1 + q + \cdots + q^{j-1})$$

is a nonconstant polynomial with nonnegative coefficients. It has one positive-base inverse for every $y > 1$; on $(0, 1)$, the range is $(1, n!)$.

10.2.2 q -multinomial coefficients

For a composition $n_1 + \cdots + n_r = n$,

$$\left[\begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \frac{[n]_q!}{[n_1]_q! \cdots [n_r]_q!}$$

is a polynomial with nonnegative coefficients. Unless the composition is trivial, every target greater than one has one positive-base inverse. Near $q = 1$, its logarithm is the cumulant-generating function of the multiset inversion statistic, so mean and variance determine the first inverse terms.

10.2.3 Rogers–Szego polynomials

For fixed $z > 0$,

$$H_n(z; q) = \sum_{k=0}^n \left[\begin{matrix} n \\ k \end{matrix} \right]_q z^k$$

has nonnegative coefficients as a polynomial in q . It therefore has a unique positive-base inverse for every target above $H_n(z; 0)$, unless $n \leq 1$ makes it constant in q .

10.2.4 Partition polynomials and selected q -Catalan numbers

Any finite partition generating polynomial $\sum_{\lambda \in \mathcal{F}} q^{w(\lambda)}$ with a nonconstant nonnegative weight has a unique positive-base inverse. The same conclusion applies to those normalizations of q -Catalan numbers known to be polynomials with nonnegative coefficients. Because several inequivalent q -Catalan families exist, positivity should be checked for the chosen definition rather than inferred from the name alone.

10.3 Probabilistic inversion

Normalize F at $q = 1$ and let X have probabilities $\Pr(X = j) = c_j / F(1)$. Then

$$\log \frac{F(e^t)}{F(1)} = \log \mathbb{E} e^{tX} = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!},$$

where κ_r are cumulants. Inverting a target near $F(1)$ is thus series reversion of a cumulant expansion. For palindromic polynomials, all odd centered cumulants beyond the first vanish, explaining the missing cubic term in the Gaussian forward expansion and the simple inverse coefficients in Equation (8.3).

Principle 10.2 (Cumulant-to-inverse pipeline). *For any positive-coefficient q -analogue with a known combinatorial statistic, compute its cumulants, form the logarithmic target displacement, and revert the cumulant series. This separates combinatorics (cumulants) from analysis (series inversion) and applies uniformly to Gaussian, multinomial, Mahonian, and partition polynomials.*

11 Inverse q -numbers and elementary deformations

11.1 The q -number

For $q > 0$, $q \neq 1$, define

$$[x]_q = \frac{1 - q^x}{1 - q}.$$

Theorem 11.1 (PAPER PROOF Exact inverse in x). *For a fixed branch of the logarithm and an admissible target,*

$$x = \frac{\text{Log}(1 - (1 - q)y)}{\text{Log } q}. \quad (11.1)$$

For real $0 < q < 1$, this gives a strictly increasing bijection $x \in (0, \infty) \mapsto y \in (0, (1 - q)^{-1})$.

Proof. Rearrange to $q^x = 1 - (1 - q)y$ and take a chosen logarithm. Real monotonicity follows from q^x decreasing in x . $\square$

As a complex inverse, (11.1) has the logarithmic lattice

$$x_m = \frac{\text{Log}(1 - (1 - q)y) + 2\pi im}{\text{Log } q}.$$

11.2 Recovering the base of a q -number

Theorem 11.2 (PAPER PROOF Real base monotonicity). *Fix $x > 0$, $x \neq 1$. On $0 < q < 1$, the function $[x]_q$ is strictly increasing from 1 to x if $x > 1$, and strictly decreasing from 1 to x if $0 < x < 1$. Thus every target strictly between 1 and x has one base inverse.*

Proof. Use the integral average

$$[x]_q = \frac{1}{1 - q} \int_q^1 x t^{x-1} dt.$$

If $x > 1$, the integrand is increasing, so deleting the lower part of the interval raises its average. If $0 < x < 1$, it is decreasing, so the average falls. The endpoint limits are immediate. $\square$

For integer $x = n$, the inverse equation is the polynomial

$$1 + q + \cdots + q^{n-1} = y.$$

The positive branch is unique, while the complex inverse has $n - 1$ sheets.

11.3 Other elementary q -deformations

The same workflow applies to symmetric q -numbers $(q^x - q^{-x})/(q - q^{-1})$, Tsallis-type logarithms, and deformed powers, but they have different branch symmetries and should not be conflated with the product-theoretic inverses developed below. In every case the correct starting point is the parameter derivative and its zeros, followed by a declared normalization at $q = 1$ or another regular base.

Part IV

Inverse q -special functions

12 The inverse q -gamma and q -polygamma functions

12.1 Two real argument branches of Γ_q

For $0 < q < 1$,

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}, \quad x > 0.$$

Its logarithmic derivative is

$$\psi_q(x) = -\log(1-q) + \log q \sum_{j=0}^{\infty} \frac{q^{x+j}}{1-q^{x+j}}, \quad (12.1)$$

and

$$\psi'_q(x) = (\log q)^2 \sum_{j=0}^{\infty} \frac{q^{x+j}}{(1-q^{x+j})^2} > 0. \quad (12.2)$$

Theorem 12.1 (PAPER PROOF Unique minimum and two inverse branches). *There is a unique $x_q^* > 0$ satisfying $\psi_q(x_q^*) = 0$. The function Γ_q decreases strictly from $+\infty$ to*

$$m_q = \Gamma_q(x_q^*)$$

on $(0, x_q^)$, then increases strictly to $+\infty$ on (x_q^*, ∞) . Thus every $y > m_q$ has exactly two positive inverse values*

$$\Gamma_{q,-}^{-1}(y) < x_q^* < \Gamma_{q,+}^{-1}(y),$$

while $y = m_q$ has one double inverse value.

Proof. Equation (12.2) makes ψ_q strictly increasing. As $x \downarrow 0$, its first summand yields $\psi_q(x) = -1/x + O(1)$, so the limit is $-\infty$. As $x \rightarrow \infty$, the sum tends to zero and $\psi_q(x) \rightarrow -\log(1-q) > 0$. Thus there is exactly one zero. Since $\Gamma'_q = \Gamma_q \psi_q$, this is the unique minimum. The product formula shows $\Gamma_q(x) \rightarrow \infty$ at both endpoints. $\square$

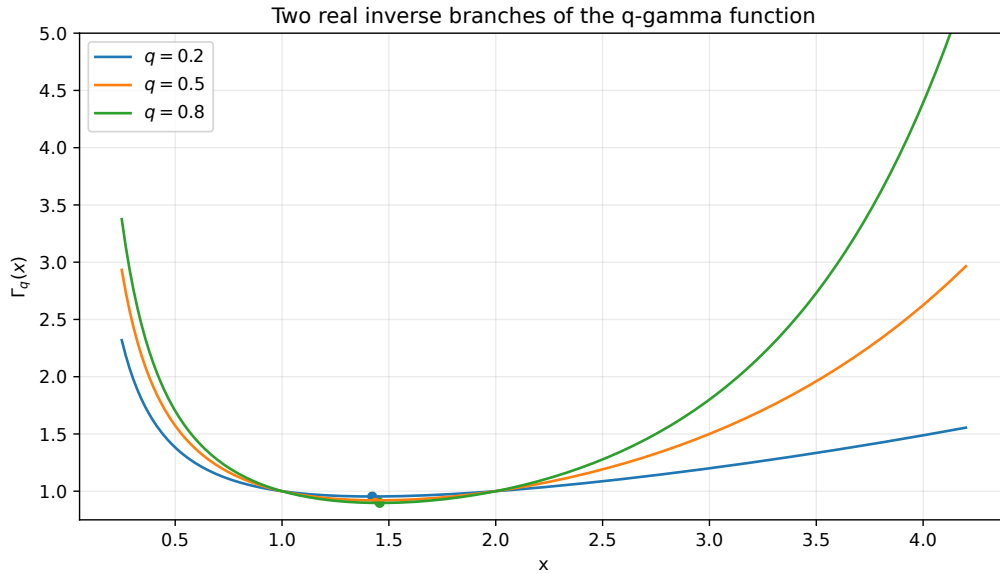


Figure 12.1: The unique minimum and the two positive inverse branches of Γ_q .

12.2 Puiseux expansion at the minimum

Let

$$\tau_q = \psi'_q(x_q^*) > 0, \quad \sigma_q = \psi''_q(x_q^*) < 0.$$

At the minimum,

$$\Gamma''_q(x_q^*) = m_q \tau_q, \quad \Gamma'''_q(x_q^*) = m_q \sigma_q.$$

Hence

$$\Gamma_{q,\pm}^{-1}(y) = x_q^* \pm \sqrt{\frac{2(y - m_q)}{m_q \tau_q}} - \frac{\sigma_q}{3m_q \tau_q^2}(y - m_q) + O((y - m_q)^{3/2}). \quad (12.3)$$

This expansion is ideal for initializing both branches when the target is close to the minimum, where direct Newton iteration is ill-conditioned.

12.3 Endpoint asymptotics of the two branches

Put

$$\lambda = -\log q, \quad L = -\log(1 - q), \quad C_q = (1 - q)(q; q)_\infty.$$

As $x \downarrow 0$,

$$\Gamma_q(x) = \frac{1 - q}{\lambda x} (1 + O(x)),$$

so

$$\Gamma_{q,-}^{-1}(y) = \frac{1 - q}{\lambda y} (1 + O(y^{-1})), \quad y \rightarrow \infty. \quad (12.4)$$

As $x \rightarrow \infty$,

$$\Gamma_q(x) = C_q e^{Lx} \left(1 + \frac{q^x}{1 - q} + O(q^{2x}) \right),$$

which yields

$$\begin{aligned} \Gamma_{q,+}^{-1}(y) &= \frac{\log(y/C_q)}{L} \\ &\quad - \frac{1}{L(1 - q)} \left(\frac{C_q}{y} \right)^{\lambda/L} + O\left(y^{-2\lambda/L} + y^{-\lambda/L-1}\right). \end{aligned} \quad (12.5)$$

The two branches therefore have radically different large-target scales: reciprocal on the left and logarithmic on the right.

12.4 Recovering the base of $\Gamma_q(x)$

The dependence on q is governed by a classical monotonicity theorem recorded in the DLMF [2].

Theorem 12.2 (Known monotonicity; [PAPER PROOF](#)). *Fix $x > 0$.*

(i) *If $0 < x < 1$ or $x > 2$, then $q \mapsto \Gamma_q(x)$ is strictly increasing on $(0, 1)$, from 1 to $\Gamma(x)$.*

(ii) *If $1 < x < 2$, it is strictly decreasing from 1 to $\Gamma(x)$.*

(iii) *At $x = 1$ and $x = 2$, it is identically one.*

Consequently the base inverse is unique whenever the target lies strictly between 1 and $\Gamma(x)$, with the order of the endpoints determined by the stated sector.

At $q \rightarrow 0^+$,

$$\log \Gamma_q(x) = q^x + (x - 2)q + o(q^x + q), \quad (12.6)$$

away from the exact exceptional points $x = 1, 2$. Therefore

$$\begin{aligned} q &\sim (y - 1)^{1/x}, & 0 < x < 1, \\ q &\sim \frac{1 - y}{2 - x}, & 1 < x < 2, \\ q &\sim \frac{y - 1}{x - 2}, & x > 2. \end{aligned}$$

At the classical endpoint, with $q = e^{-t}$,

$$\log \Gamma_q(x) = \log \Gamma(x) - \frac{(x-1)(x-2)}{4}t + O(t^2). \quad (12.7)$$

Thus, if $y \rightarrow \Gamma(x)$ within the appropriate monotonic range,

$$-\log q = \frac{4 \log(\Gamma(x)/y)}{(x-1)(x-2)} + O(\log(\Gamma(x)/y)^2). \quad (12.8)$$

The quotient is positive in all three nonconstant sectors.

12.5 The minimizer as a function of the base

The numerical data in the archive show

$$x_{0.05}^* = 1.37998346149 \dots, \quad x_{0.95}^* = 1.46060675866 \dots,$$

while the classical gamma minimizer is $x^* = 1.461632144968 \dots$

For $q \rightarrow 0^+$, let $\lambda = -\log q$. Balancing the leading terms in $\psi_q(x) = 0$, namely $q - \lambda q^x = 0$, predicts

$$x_q^* = 1 + \frac{\log \lambda}{\lambda} + o(\lambda^{-1}). \quad (12.9)$$

For $q \rightarrow 1^-$, local uniform convergence of ψ_q to the classical ψ implies $x_q^* \rightarrow x^*$.

Conjecture 12.3 (**CONJECTURAL** Monotone minimizer). *The function $q \mapsto x_q^*$ is strictly increasing on $(0, 1)$, and the minimum value m_q is strictly decreasing from 1 to $\Gamma(x^*)$.*

The data in `data/qgamma_minima.csv` support both assertions. A proof should follow from a sign analysis of

$$\frac{dx_q^*}{dq} = - \frac{\partial_q \psi_q(x_q^*)}{\psi_q'(x_q^*)},$$

so the problem reduces to the sign of one explicit Lambert-series derivative.

12.6 Inverse q -digamma and q -polygamma functions

Expanding the geometric denominator gives

$$\psi_q(x) = -\log(1-q) + \log q \sum_{\ell=1}^{\infty} \frac{q^{\ell x}}{1-q^{\ell}}, \quad (12.10)$$

and, for $m \geq 1$,

$$\psi_q^{(m)}(x) = (\log q)^{m+1} \sum_{\ell=1}^{\infty} \frac{\ell^m q^{\ell x}}{1-q^{\ell}}. \quad (12.11)$$

Theorem 12.4 (**PAPER PROOF** Global inverse polygamma branches). *For fixed $0 < q < 1$:*

(i) ψ_q is a strictly increasing bijection from $(0, \infty)$ onto $(-\infty, -\log(1-q))$. It has one real inverse for every target in this range.

(ii) For every $m \geq 1$, the positive function

$$g_{q,m}(x) = (-1)^{m+1} \psi_q^{(m)}(x)$$

is strictly decreasing from $+\infty$ to 0. Hence it has one real inverse for every positive target.

Proof. The first assertion follows from $\psi_q' > 0$ and its endpoint limits. Formula (12.11) shows $g_{q,m} > 0$; differentiating gives $g_{q,m}' = -g_{q,m+1} < 0$. The endpoint limits follow from the first terms and comparison with the corresponding powers of x^{-1} . $\square$

This family provides unusually clean examples of globally invertible transcendental q -functions.

13 The inverse q -beta function

13.1 Coordinate inverses

For $a, b > 0$,

$$B_q(a, b) = \frac{\Gamma_q(a)\Gamma_q(b)}{\Gamma_q(a+b)}.$$

Theorem 13.1 (PAPER PROOF Global inverse in either shape parameter). *Fix $0 < q < 1$ and $b > 0$. The function $a \mapsto B_q(a, b)$ is strictly decreasing and log-convex on $(0, \infty)$, from $+\infty$ to*

$$\ell_{q,b} = (1-q)^b \Gamma_q(b). \quad (13.1)$$

Therefore every $y > \ell_{q,b}$ has exactly one inverse a . The same statement holds with a and b interchanged.

Proof. Differentiation gives

$$\partial_a \log B_q(a, b) = \psi_q(a) - \psi_q(a+b) < 0,$$

and

$$\partial_a^2 \log B_q(a, b) = \psi'_q(a) - \psi'_q(a+b) > 0$$

because ψ_q increases and ψ'_q decreases. The limit at zero is infinite because $\Gamma_q(a)$ has a simple pole. The product formula gives

$$B_q(a, b) = (1-q)^b \Gamma_q(b) \frac{(q^{a+b}; q)_\infty}{(q^a; q)_\infty} = \frac{\ell_{q,b}}{(q^a; q)_b},$$

which tends to $\ell_{q,b}$ as $a \rightarrow \infty$. □

The Hessian of $\log B_q$ is

$$\begin{pmatrix} \psi'_q(a) - \psi'_q(a+b) & -\psi'_q(a+b) \\ -\psi'_q(a+b) & \psi'_q(b) - \psi'_q(a+b) \end{pmatrix}.$$

Coordinate convexity is immediate; joint convexity requires positivity of the determinant and is a separate question.

13.2 Endpoint inverse asymptotics

As $a \downarrow 0$,

$$B_q(a, b) = \frac{1-q}{(-\log q)a} (1 + O(a)),$$

so

$$a(y) = \frac{1-q}{(-\log q)y} (1 + O(y^{-1})), \quad y \rightarrow \infty. \quad (13.2)$$

At the other endpoint,

$$B_q(a, b) = \ell_{q,b} (1 + [b]_q q^a + O(q^{2a})),$$

and therefore

$$a(y) = \frac{\log((y - \ell_{q,b})/(\ell_{q,b}[b]_q))}{\log q} + o(1), \quad y \downarrow \ell_{q,b}. \quad (13.3)$$

Again, approach to a finite floor produces logarithmic divergence of the inverse parameter.

13.3 Inversion with respect to q

The product representation

$$B_q(a, b) = (1 - q) \frac{(q; q)_\infty (q^{a+b}; q)_\infty}{(q^a; q)_\infty (q^b; q)_\infty}$$

gives the exact kernel

$$\begin{aligned} q \partial_q \log B_q(a, b) = & -\frac{q}{1-q} - \sum_{j=1}^{\infty} \frac{j q^j}{1-q^j} - \sum_{j=0}^{\infty} \frac{(a+b+j) q^{a+b+j}}{1-q^{a+b+j}} \\ & + \sum_{j=0}^{\infty} \frac{(a+j) q^{a+j}}{1-q^{a+j}} + \sum_{j=0}^{\infty} \frac{(b+j) q^{b+j}}{1-q^{b+j}}. \end{aligned} \quad (13.4)$$

Unlike the coordinate derivatives, its sign is not transparent for arbitrary real a, b . The endpoint values are

$$\lim_{q \rightarrow 0^+} B_q(a, b) = 1, \quad \lim_{q \rightarrow 1^-} B_q(a, b) = B(a, b).$$

Depending on whether the classical beta value is above or below one, an increasing or decreasing trend is possible; internal extrema are not excluded without a sign theorem for (13.4).

For positive integer parameters, however, the answer is complete.

Theorem 13.2 (**PAPER PROOF** Integer-parameter base inverse). *Let $m, n \in \mathbb{N}$, not both equal to one. Then*

$$B_q(m, n) = \frac{1}{[n]_q \begin{bmatrix} m+n-1 \\ m-1 \end{bmatrix}_q}. \quad (13.5)$$

It is strictly decreasing for $q > 0$, from 1 to 0. Hence every $y \in (0, 1)$ has exactly one positive-base inverse. Restricted to $0 < q < 1$, the target range is $(B(m, n), 1)$.

Proof. Use $\Gamma_q(r) = [r-1]_q!$ and simplify to (13.5). The denominator is a nonconstant product of positive-coefficient polynomials and is strictly increasing for $q > 0$. $\square$

Conjecture 13.3 (**CONJECTURAL** Real-parameter beta sign regions). *There is a finite collection of explicit curves in the positive (a, b) -quadrant separating regions on which $q \mapsto B_q(a, b)$ is monotone on $(0, 1)$. In particular, it is expected to be decreasing throughout $a \geq 1, b \geq 1$, extending the integer theorem.*

A proof or counterexample should begin with complete-monotonicity transforms of (13.4), not with direct numerical differentiation.

14 Euler q -exponentials and product logarithms

14.1 Exact inverse reductions

Use the unscaled Euler conventions

$$e_q(z) = \frac{1}{(z; q)_\infty}, \quad E_q(z) = (-z; q)_\infty, \quad 0 < q < 1.$$

(The common scaled conventions replace z by $(1-q)z$.)

Theorem 14.1 (PAPER PROOF Exact real inverses). *On their principal real branches,*

$$e_q^{-1}(y) = \mathcal{A}_q(1/y), \quad y > 0, \quad (14.1)$$

$$E_q^{-1}(y) = -\mathcal{A}_q(y), \quad y > 0. \quad (14.2)$$

For the scaled functions $\tilde{e}_q(z) = 1/((1-q)z; q)_\infty$ and $\tilde{E}_q(z) = (-(1-q)z; q)_\infty$, divide the right sides by $1-q$.

Proof. Equations (14.1)–(14.2) are immediate from the definitions and the uniqueness of the principal product inverse. $\square$

This suggests product-theoretic q -logarithms

$$\log_q^{(e)}(y) = \mathcal{A}_q(1/y), \quad \log_q^{(E)}(y) = -\mathcal{A}_q(y).$$

They are distinct from Tsallis and Jackson logarithms. Their analytic theory is inherited exactly from $\mathcal{A}_q$, including the Bernstein conjecture, branch points, and endpoint expansions.

14.2 Asymptotic consequences

From (5.4), as $y \rightarrow \infty$,

$$e_q^{-1}(y) = 1 - \frac{1}{(q; q)_\infty y} + \frac{L_q}{(q; q)_\infty^2 y^2} + O(y^{-3}).$$

From (5.6), as $y \rightarrow \infty$,

$$\log E_q^{-1}(y) = \sqrt{2(-\log q) \log y} + \frac{1}{2} \log q + O((\log y)^{-1/2}).$$

Thus the two Euler exponentials have inverse scales approaching a finite singularity and growing stretched-exponentially, respectively.

14.3 Inverse semigroup questions

The ordinary logarithm linearizes multiplication. Neither product logarithm above does so exactly. One may instead define a deformed operation

$$x \oplus_q y = e_q^{-1}(e_q(x)e_q(y)),$$

on a branch where the product remains admissible. Associativity follows from ordinary multiplication, while the explicit operation is expressed through $\mathcal{A}_q$. Its Taylor coefficients, singularities, and classical limit form a concrete inverse-function approach to q -addition.

Research problem 14.2. Classify the real and complex Lie-semigroup structure induced by $\oplus_q$, determine whether its infinitesimal generator has a simple Lambert-series representation, and compare it with Jackson's and Tsallis' standard deformed additions.

15 Basic hypergeometric, theta, and orthogonal-polynomial inverses

15.1 Local inverse of a basic hypergeometric series

In a standard convention,

$${}_r\phi_s \left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right) = \sum_{m \geq 0} c_m z^m,$$

where

$$c_m = \frac{(a_1, \dots, a_r; q)_m}{(q, b_1, \dots, b_s; q)_m} \left((-1)^m q^{\binom{m}{2}} \right)^{1+s-r}.$$

Assume $c_1 \neq 0$. Since the value at $z = 0$ is one, the local inverse in the argument is

$$z = \frac{y-1}{c_1} - \frac{c_2}{c_1^3}(y-1)^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) (y-1)^3 + O((y-1)^4). \quad (15.1)$$

If $c_1 = 0$ but $c_r \neq 0$ is the first nonzero coefficient, the inverse has r -th-root ramification at $y = 1$.

Parameter inverses are obtained from the same implicit formula

$$\frac{\partial p}{\partial y} = \frac{1}{F_p}, \quad \frac{\partial p}{\partial a_i} = -\frac{F_{a_i}}{F_p},$$

where every parameter derivative of a coefficient reduces to finite or infinite q -Pochhammer kernels. Singular divisors occur when denominator factors vanish, and branch points occur when the chosen parameter derivative vanishes.

15.2 Theta-function inversion

Define

$$\theta(z; q) = (z; q)_\infty (q/z; q)_\infty.$$

It satisfies

$$\theta(qz; q) = -z^{-1} \theta(z; q)$$

and has zeros at $z = q^m$, $m \in \mathbb{Z}$. Its logarithmic argument derivative is

$$z \partial_z \log \theta(z; q) = - \sum_{j \geq 0} \frac{z q^j}{1 - z q^j} + \sum_{j \geq 0} \frac{q^{j+1}/z}{1 - q^{j+1}/z}. \quad (15.2)$$

Thus inverse theta branch points are the images of zeros of the balanced Lambert kernel (15.2). Quasi-periodicity transports local data between annuli, while the two product factors make the inverse surface doubly infinite.

The phase function in (5.5) is itself a theta-product factor. Large-target inverse q -Pochhammer asymptotics and theta inversion are therefore two aspects of the same multiplicative lattice geometry.

15.3 q -orthogonal polynomials

For fixed degree and parameters, an orthogonal polynomial $p_n(x; q)$ has an algebraic inverse in x . Its real branches are divided by its critical points, which interlace the zeros under the usual positive orthogonality hypotheses. The branch-value polynomial is

$$\text{Disc}_x(p_n(x; q) - y).$$

Inversion in a shape parameter or in q is usually transcendental even when inversion in x is algebraic. Parameter derivatives can be obtained from three-term recurrences by differentiating the recurrence and propagating both p_n and its sensitivity.

Principle 15.1 (Recurrence-assisted inverse continuation). *For a family satisfying*

$$p_{n+1}(x) = (A_n x + B_n) p_n(x) - C_n p_{n-1}(x),$$

propagate p_n , $\partial_x p_n$, and all needed parameter derivatives through differentiated recurrences. Use these quantities in a branch-aware Newton corrector. This avoids unstable coefficient expansion at large degree.

15.4 q -Bessel and q -Airy functions

In standard parameter regimes, Jackson q -Bessel and q -Airy functions may exhibit infinitely many zeros and critical values. When that structure is present, their argument inverses are naturally indexed by zero intervals on the real axis and by continuation paths in the complex plane. As with the ordinary Bessel inverse, a single principal branch captures only a small part of the global surface. Uniform asymptotics for large zero index can be inverted to initialize remote branches; near coalescing turning points, Airy-type uniformization replaces ordinary Taylor reversion.

Research problem 15.2. Construct a DLMF-style branch atlas for inverse Jackson q -Bessel and q -Airy functions, including zero-index labels, critical values, $q \rightarrow 1$ transition formulas, and certified continuation algorithms. The product-kernel methods of this report supply the parameter derivatives needed for such an atlas.

Part V

Computation, certification, and research frontiers

16 Branch-aware numerical inversion

16.1 Why unconstrained Newton iteration is insufficient

Newton's update for $F(p) = y$ is

$$p_{m+1} = p_m - \frac{F(p_m) - y}{F'(p_m)}.$$

Near a simple root it converges quadratically, but globally it can cross poles, jump over critical points, or converge to a different sheet. A reliable inverse routine must carry a branch certificate: a monotone interval on the real axis, or a continuation path and sheet label in the complex plane.

16.2 Bracketed Newton on a real branch

Algorithm 16.1 (PAPER PROOF Safeguarded branch inversion). Suppose $F \in C^2([L, R])$ is strictly monotone, $|F'(p)| \geq m > 0$ throughout the interval, and y lies between $F(L)$ and $F(R)$.

1. Maintain a bracket $[L_m, R_m]$ known to contain the unique inverse.
2. Propose the Newton point

$$N_m = p_m - \frac{F(p_m) - y}{F'(p_m)}.$$

3. Accept N_m only if it lies in the middle half $[L_m + (R_m - L_m)/4, R_m - (R_m - L_m)/4]$; otherwise replace it by the midpoint.
4. Update the bracket using monotonicity, set $p_{m+1} = N_m$, and stop when both the residual and bracket width meet tolerance.

The bracket never loses the solution and its width contracts by at least a factor $3/4$ per step. Once unsafeguarded Newton points remain accepted in the local basin, the iterates converge quadratically.

This algorithm is used in the accompanying code for the principal q -Pochhammer inverse, finite and infinite critical points, Gaussian base inverses, and the two q -gamma branches.

16.3 Stable product evaluation

For positive real factors, evaluate

$$\log(a; q)_\infty = \sum_{j \geq 0} \log(1 - aq^j)$$

rather than multiplying directly. Use $\log 1p$ for small aq^j , and truncate only after applying the tail bound of [Lemma 3.3](#). For signed real intervals, separate the sign of the finite number of negative factors and accumulate the logarithm of the absolute value.

For derivatives, logarithmic kernels are usually more stable than direct finite differences. At a principal inverse point $a = \mathcal{A}_q(y)$, for example,

$$\mathcal{A}'_q(y) = -\frac{1}{yS_1(a)}$$

requires only a positive Lambert sum.

16.4 Complex predictor–corrector continuation

Suppose a branch value $p(y_0)$ is known and a path $y(s)$, $0 \leq s \leq 1$, avoids critical values. Differentiate $F(p(s)) = y(s)$:

$$p'(s) = \frac{y'(s)}{F'(p(s))}.$$

A first-order predictor is

$$p_{\text{pred}} = p_m + \frac{\Delta y}{F'(p_m)}.$$

Correct it with Newton’s method at the new target. Step size should be reduced when $|F'|$ becomes small or the corrector takes too many iterations. Near a known branch point, switch to the local square-root coordinate $w = \sqrt{y - y_c}$, in which the sheets become analytic.

Algorithm 16.2 (Complex inverse continuation). 1. Choose a base point, a branch root, and a target path avoiding the critical-value set.

2. Predict using the implicit differential equation.
3. Correct with Newton iteration and verify the residual.
4. Compare the corrected root with all nearby known roots to detect an unintended sheet jump.
5. Adapt the step according to $|F'|$, Newton contraction, and distance to known critical values.
6. Record the permutation after closed loops to compute monodromy.

16.5 Locating branch points

For a one-variable direct map, solve

$$F_p(p; \lambda) = 0, \quad y = F(p; \lambda).$$

For finite algebraic families, eliminate p using a resultant or discriminant. To find parameter values where branch values collide, compute a second discriminant in y . This is exactly how [\(4.8\)](#) was found.

For transcendental products, bracket the real critical point using poles of the logarithmic derivative. In the complex plane, use argument-principle counts to ensure that all critical points in a box or disk have been found:

$$N = \frac{1}{2\pi i} \int_\gamma \frac{F_{pp}(p)}{F_p(p)} dp.$$

Subdivide until each region contains one root, then apply interval Newton or ball arithmetic.

16.6 Formal and interval certification

The following hierarchy separates levels of trust.

- (i) *Floating-point evidence*: useful for discovery, not proof.
- (ii) *Arbitrary-precision residual*: rules out ordinary roundoff but not a systematic branch or truncation error.
- (iii) *Interval or ball proof*: encloses products, derivatives, and roots with directed rounding.
- (iv) *Exact algebra*: proves polynomial identities and isolating intervals using rational arithmetic and Sturm sequences.
- (v) *Proof-assistant formalization*: verifies definitions, inequalities, and algorithms inside a foundational kernel.

The secondary-discriminant factorizations in the archive are exact algebra. The decimal identification of colliding critical intervals is high-precision numerics and is explicitly labelled as such.

16.7 Complexity considerations

A truncated infinite product with N factors costs $O(N)$. As $q \rightarrow 1$, the tail bound in [Lemma 3.3](#) gives, for fixed nonzero a ,

$$N = \Theta\left(\frac{\log(1/\varepsilon) + \log(1/(1-q))}{1-q}\right)$$

in the naive fixed-tolerance regime. Polylogarithmic asymptotics or modular transformations therefore become essential. Finite Gaussian coefficients should be evaluated through logarithmic products or recurrences rather than expanded coefficients when n is large.

For many targets at fixed parameters, precompute critical points and build a branch atlas. Chebyshev or rational approximants on each monotone interval can then reduce repeated inversion to a few Newton corrections.

17 Reproducible experiments

17.1 Archive layout

The companion program is `numerical_experiments.py`. It uses only `mpmath`, `sympy`, `numpy`, and `matplotlib`, and it creates:

- exact secondary discriminants for $3 \leq n \leq 6$;
- finite critical points and values for several n, q ;
- infinite critical-point asymptotic ratios;
- Gaussian inverse errors near $q = 1$;
- q -gamma minimizers;
- inverse-product derivatives through order eight;
- all figures included in this report.

No external network calls are made.

17.2 Experiment A: the first critical-value collision

The symbolic calculation forms

$$P_n(a; q) = \prod_{j=0}^{n-1} (1 - aq^j), \quad \Delta_n(y; q) = \text{Disc}_a(P_n - y), \quad \mathfrak{M}_n(q) = \text{Disc}_y \Delta_n.$$

Factoring over $\mathbb{Z}[q]$ produces (4.5)–(4.8). Bisection on the exact factor H_5 gives the quoted q_0 . Independent bracketed solutions of the logarithmic critical equation locate $c_{5,0}$ and $c_{5,2}$.

A useful lesson is methodological: a plausible numerical conjecture that critical-value magnitudes increase with the interval index is false near $q = 1$. Exact elimination revealed the first crossing and replaced the false conjecture with a precise Maxwell-locus theory.

17.3 Experiment B: remote critical asymptotics

For each $q \in \{0.35, 0.55, 0.75\}$ and $1 \leq k \leq 14$, the program solves

$$\sum_{j \geq 0} \frac{q^j}{1 - aq^j} = 0$$

on $(q^{-k}, q^{-(k+1)})$. It compares the result with (5.7) and (5.8). At $q = 0.35$, for example, the critical-point ratio “exact/asymptotic” improves from about 1.0388 at $k = 2$ to 1.00056 at $k = 14$, while the critical-value ratio improves from about 0.954 to 0.996.

The convergence slows as $q \rightarrow 1$, which is consistent with loss of uniformity when the product tail length grows. A double asymptotic in k and $1 - q$ is therefore a natural next step.

17.4 Experiment C: inverse Bernoulli expansion

For $n = 12$, $k = 5$, and targets $y = \binom{12}{5} e^{-s}$, the exact positive base is found by bisection. At $s = 10^{-4}$, the cubic approximation in (8.3) has relative error below 10^{-15} in $t = -\log q$, whereas the quadratic error is about 7.7×10^{-11} . The log-log slopes in Figure 8.1 exhibit the predicted orders.

17.5 Experiment D: the q -gamma minimum

The root of $\psi_q(x) = 0$ is found by safeguarded Newton iteration using the positive trigamma derivative. Selected values are

q	x_q^*	m_q
0.05	1.37998346149	0.98270463141
0.20	1.42196580342	0.95376342969
0.60	1.45066979499	0.91169489999
0.95	1.46060675866	0.88840573828

The trend supports the monotone-minimizer conjecture and approaches the classical gamma minimum.

17.6 Experiment E: higher derivatives of the principal inverse

At a chosen direct argument $a < 1$, write

$$(a + h; q)_\infty = (a; q)_\infty \prod_{j \geq 0} (1 - hw_j), \quad w_j = \frac{q^j}{1 - aq^j}.$$

Newton identities recover the Taylor coefficients from the power sums $\sum_j w_j^r$, avoiding unstable numerical differentiation. Reverting this series through degree eight tests the signs of inverse derivatives. All archived cases satisfy

$$(-1)^r \mathcal{A}_q^{(r)}(y) > 0, \quad 1 \leq r \leq 8,$$

which is equivalent to the first eight Bernstein sign conditions for $1 - \mathcal{A}_q$.

17.7 Limitations of the experiments

The calculations do not prove the Bernstein conjecture, monotonicity of the q -gamma minimizer, or the proposed real-parameter beta sign regions. The code uses high precision and analytic bracketing, but not directed-rounding interval arithmetic. Its purpose is reproducible discovery and stress testing of formulas. Every conjectural statement remains labelled.

18 Research program and conjectures

18.1 Finite-product Maxwell geometry

Conjecture 18.1 (**CONJECTURAL** Structured critical-value crossings). *For fixed n , the positive real zeros of the reduced secondary discriminant $\mathfrak{M}_n(q)$, after removing powers of q , $q - 1$, and cyclotomic factors, are simple at the level of the collision parameter and correspond to pairwise equalities of same-sign real critical values. The number and ordering of these roots admit a finite combinatorial description in terms of interval indices.*

Questions include:

- (a) Does the first collision always involve I_0 and I_2 for $n \geq 5$?
- (b) How many positive Maxwell parameters occur as $n \rightarrow \infty$?
- (c) Do reciprocal collision factors always reduce to polynomials in $q + q^{-1}$?
- (d) What is the limiting distribution of their roots near $q = 1$?

Exact resultant computations for moderate n , followed by asymptotic analysis of critical intervals, form a realistic attack.

18.2 A full expansion of remote infinite-product extrema

Conjecture 18.2 (**CONJECTURAL** Complete $1/k$ expansion). *For fixed $0 < q < 1$, the critical-point scale in (5.9) admits expansions*

$$\begin{aligned} \varepsilon_k &\sim \frac{1}{k+2} + \sum_{r \geq 2} \frac{E_r(q)}{(k+2)^r}, \\ c_k &\sim q^{-(k+1)} \left(1 - \sum_{r \geq 1} \frac{C_r(q)}{(k+2)^r} \right), \\ \frac{|(c_k; q)_\infty|}{(q; q)_\infty^2 q^{-(k+1)(k+2)/2}} &\sim \frac{e^{-1}}{k+2} \left(1 + \sum_{r \geq 1} \frac{V_r(q)}{(k+2)^r} \right), \end{aligned}$$

where the coefficients are explicit polynomials in Lambert series such as $\sum_{m \geq 1} q^m / (1 - q^m)^r$.

The proof should expand $\partial_\varepsilon \log G_m$ uniformly and apply implicit series reversion. Exponentially small q^m terms can then be added as a transseries correction.

18.3 Double scaling and Lambert W

Fixed- a inversion near $q = 1$ is controlled by $\text{Li}_2(a)/(-\log q)$. When $a \rightarrow 1$ simultaneously, the dilogarithm and $\log(1 - a)$ terms compete. Scaling laws such as

$$1 - a \asymp -\log q, \quad -\log y \asymp \frac{1}{-\log q},$$

lead to equations in which the unknown appears both linearly and inside a logarithm or exponential. Lambert W , including its W_{-1} branch, is then expected to provide uniform transition coordinates.

Research problem 18.3. Derive uniform inverse formulas for $(a; q)_\infty = y$ in all joint regimes $(a, q, y) \rightarrow (1, 1, 0)$. Identify the transition surfaces between algebraic, logarithmic, dilogarithmic, and Lambert- W scales, and prove uniform error bounds.

18.4 Complete Bernstein and Pick geometry

A proof that $1 - \mathcal{A}_q$ is complete Bernstein would imply a representation

$$1 - \mathcal{A}_q(y) = c_0 + c_1 y + \int_0^\infty \frac{y}{y+t} \mu_q(dt)$$

for a positive measure μ_q . The singular support should encode the negative or complex critical values of the direct product. The challenge is that inverse-function singularities are not simply the original zero set.

Research problem 18.4. Construct, or rule out, a Stieltjes measure for $(1 - \mathcal{A}_q(y))/y$. Relate its moments to the inverse coefficients at $y = 0$ or $y = 1$, and determine whether finite products provide monotone approximants in the complete Bernstein cone.

18.5 Galois and monodromy theory

For finite n , consider

$$P_n(a; q) - y \in \mathbb{Q}(q, y)[a].$$

Away from critical-value collisions, the analytic monodromy is S_n , which strongly suggests the generic algebraic Galois group over $\mathbb{Q}(q, y)$ is also S_n .

Conjecture 18.5 (**CONJECTURAL** Generic Galois group). *For every $n \geq 2$, the polynomial $(a; q)_n - y$ has Galois group S_n over $\mathbb{Q}(q, y)$. Specializations with smaller group are contained in an explicit thin set controlled by discriminants and resolvents.*

A proof can combine irreducibility over the rational function field, specialization at a convenient rational base, and cycle information from local ramification.

18.6 Inverse-coefficient positivity

The local inverse of a positive-coefficient polynomial need not itself have positive coefficients after arbitrary normalization. Yet the Gaussian inverse around $q = 0$ exhibits structured integer combinations of partition numbers, and the principal product inverse exhibits persistent alternating derivative signs.

Research problem 18.6. Find combinatorial interpretations for the coefficients of $\mathcal{Q}_{n,k}(1 + u)$ and $1 - \mathcal{A}_q(y)$. Determine whether sign-regularity, total positivity, or continued-fraction expansions follow from the underlying partition posets.

18.7 Inverse generalized Gaussian monotonicity

The integer positive-base theorem is complete, while the continuously interpolated $\mathcal{G}_q(\alpha, \beta)$ still lacks a global base-sign diagram.

Conjecture 18.7 (CONJECTURAL). *For $\alpha > \beta > 0$, $q \mapsto \mathcal{G}_q(\alpha, \beta)$ is strictly increasing on $(0, 1)$, with the boundary cases $\beta = 0$ and $\beta = \alpha$ constant. If this is false, the counterexample set is separated from the integer lattice by an explicit curve and is detected by one sign change of the base derivative.*

This is intentionally phrased to invite falsification: high-precision scans should precede any attempted proof.

18.8 Inverse q -gamma minimizer

Beyond monotonicity of x_q^* , one can ask for a complete asymptotic expansion as $q \rightarrow 0$:

$$x_q^* = 1 + \frac{\log \lambda}{\lambda} + \frac{A_2(\log \lambda)}{\lambda^2} + \cdots, \quad \lambda = -\log q.$$

Near $q = 1$, implicit differentiation of $\psi_q(x_q^*) = 0$, together with Lambert-series asymptotics, should give an ordinary power series in $-\log q$. Matching these regimes may yield accurate global rational approximants for the branch-separating minimum.

18.9 Multivariate inverse systems

Many natural identification problems contain enough observables to recover several parameters, for example

$$(a; q)_\infty = y_1, \quad \partial_a \log(a; q)_\infty = y_2.$$

The Jacobian is built from S_1, S_2 , and on $a < 1$ its sign can often be controlled. Similar systems can recover (q, x) from $(\Gamma_q(x), \psi_q(x))$.

Research problem 18.8. Classify minimally sufficient collections of q -observables whose parameter map is globally injective on natural real domains. Develop condition numbers, optimal experiment design, and certified multivariate inversion.

18.10 Formalization roadmap

A proof-assistant development can proceed in layers:

1. finite products, derivative identities, and real interval signs;
2. uniqueness of finite critical points via strict monotonicity of S_1 ;
3. exact discriminant (4.3);
4. positive-coefficient inverse theorem for Gaussian coefficients;
5. infinite products by uniform convergence and explicit tail bounds;
6. the principal inverse through a formal inverse-function theorem;
7. asymptotic critical points using quantified estimates on G_m ;
8. interval certification of the $n = 5$ collision sheet labels.

The first four layers are largely algebraic and order-theoretic and should be well suited to Lean, Isabelle, or Coq. The asymptotic layers require a mature library for infinite products and uniform big- O estimates.

19 Conclusion

The inverse theory of q -analogues is not a collection of isolated numerical root-finding tasks. It is a coherent subject organized by four structures:

- (i) logarithmic derivative kernels, which control monotonicity and conditioning;
- (ii) discriminants and critical values, which control algebraic and analytic branching;
- (iii) asymptotic reversion, which converts forward q -series expansions into inverse expansions;
- (iv) positivity and combinatorics, which can turn a many-sheeted complex problem into a unique global positive real branch.

For q -Pochhammer products, every natural parameter now has a concrete inverse framework. The finite argument inverse has a complete real atlas and an exact secondary-discriminant mechanism. The infinite argument inverse has a canonical principal branch, endpoint expansions, log-periodic large-target behavior, and an asymptotic atlas of all remote branch points. The base inverse is globally monotone in the principal sign regimes and has polylogarithmic $q \rightarrow 1$ reversion. The order inverse reduces exactly to the principal argument inverse plus a logarithmic lattice.

For Gaussian coefficients, nonnegative coefficients yield a particularly strong global theorem: every target above one determines exactly one positive base. Bernoulli and cumulant expansions then produce systematic local inverse series. Continuous parameter inverses inherit monotonicity and branch geometry from q -gamma. The same ideas extend to q -factorials, multinomials, q -beta, Euler exponentials, and broad basic hypergeometric families.

The most striking structural observation in this report is the appearance of Maxwell loci: distinct simple critical points can share a critical value as the base changes. The exact $n = 5$ collision shows that global inverse geometry cannot be inferred from local monotonicity or from a presumed ordering of extrema. Secondary discriminants provide the right organizing invariant.

The next decisive advances would be a proof of the complete Bernstein conjecture for $1 - \mathcal{A}_q$, a classification of finite-product Maxwell loci, and uniform double-scaling inversions near $(a, q) = (1, 1)$. Each is accessible to a combination of product identities, Lambert-series estimates, exact algebra, and certified computation.

A Inverse-calculus formulas

A.1 Derivatives of a one-variable inverse

Let $y = F(x)$, let $x = G(y)$, and write $f_j = F^{(j)}(x)$. At every regular point $f_1 \neq 0$, repeated implicit differentiation gives

$$G' = \frac{1}{f_1}, \tag{A.1}$$

$$G'' = -\frac{f_2}{f_1^3}, \tag{A.2}$$

$$G''' = \frac{3f_2^2 - f_1 f_3}{f_1^5}, \tag{A.3}$$

$$G^{(4)} = \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{f_1^7}, \tag{A.4}$$

$$G^{(5)} = \frac{105f_2^4 - 105f_1 f_2^2 f_3 + 10f_1^2 f_3^2 + 15f_1^2 f_2 f_4 - f_1^3 f_5}{f_1^9}. \tag{A.5}$$

All derivatives on the right are evaluated at $x = G(y)$. These formulas are especially effective for the principal inverse $\mathcal{A}_q$, because the alternating signs of the direct-product derivatives reduce sign questions to explicit Bell-polynomial inequalities.

For arbitrary order, Faà di Bruno's formula applied to $F(G(y)) = y$ yields the recursion

$$G^{(n)}(y) = -\frac{1}{F'(G(y))} \sum_{k=2}^n F^{(k)}(G(y)) B_{n,k}(G'(y), G''(y), \dots, G^{(n-k+1)}(y)), \quad (\text{A.6})$$

where $B_{n,k}$ is the partial exponential Bell polynomial. Thus every higher inverse derivative is computable without differentiating an implicit root solver.

A.2 Ordinary power-series reversion

Suppose

$$f(x) = \sum_{m \geq 1} f_m x^m, \quad f_1 \neq 0,$$

and let $g(z) = \sum_{n \geq 1} g_n z^n$ satisfy $f(g(z)) = z$. The Lagrange–Bürmann formula is

$$g_n = \frac{1}{n} [x^{n-1}] \left(\frac{x}{f(x)} \right)^n. \quad (\text{A.7})$$

For symbolic computation, a triangular recurrence is often faster:

$$g_1 = \frac{1}{f_1}, \quad g_n = -\frac{1}{f_1} [z^n] \sum_{m=2}^n f_m \left(\sum_{j=1}^{n-1} g_j z^j \right)^m, \quad n \geq 2. \quad (\text{A.8})$$

Only previously known coefficients occur on the right. Formula (A.8) is the all-order engine behind the local inverse expansions in this report.

A.3 Puiseux reversion at a simple fold

Let $F'(c) = 0$, $F''(c) \neq 0$, and $v = F(c)$. Put $\eta = y - v$ and $h = x - c$. Expanding and reverting gives the two local branches

$$h = \sigma \sqrt{\frac{2\eta}{F''(c)}} - \frac{F'''(c)}{3F''(c)^2} \eta + \sigma \frac{5F'''(c)^2 - 3F''(c)F^{(4)}(c)}{18\sqrt{2}F''(c)^{7/2}} \eta^{3/2} + O(\eta^2), \quad \sigma \in \{+1, -1\}, \quad (\text{A.9})$$

with coherent choices of square roots. A simple critical value therefore produces quadratic ramification and a transposition in local monodromy. At a Maxwell point, two such folds share a target value but remain distinct source points; at a higher caustic, one also has $F''(c) = 0$, and the Puiseux exponent changes.

A.4 Multivariate implicit inversion

For $F : \mathbb{C}^m \rightarrow \mathbb{C}^m$ and $F(p_0) = y_0$, local inversion is possible exactly when $J_F(p_0) = \det DF(p_0) \neq 0$. The first derivative of the inverse is

$$D(F^{-1})(y_0) = DF(p_0)^{-1}.$$

For a scalar observable $F(p, \lambda) = y$, with $F_p \neq 0$, variation of an auxiliary parameter is governed by

$$\frac{\partial p}{\partial \lambda} = -\frac{F_\lambda}{F_p}. \quad (\text{A.10})$$

This identity unifies the sign calculations for inverse base, order, and shape parameters. Near a branch locus the denominator warns of ill conditioning before a numerical solver visibly fails.

B Exact secondary discriminants for finite products

For

$$P_n(a; q) = \prod_{j=0}^{n-1} (1 - aq^j), \quad D_n(y; q) = \text{Disc}_a(P_n(a; q) - y),$$

define the secondary discriminant

$$\mathfrak{M}_n(q) = \text{Disc}_y D_n(y; q).$$

It vanishes when two critical values coincide, when a critical point becomes multiple, or when degree drops under a singular specialization. Cyclotomic and endpoint factors should therefore be separated from the factors that encode genuine collisions of distinct finite critical values.

The exact factorizations produced by the accompanying symbolic computation are as follows.

$$\mathfrak{M}_3(q) = 16q^6(q-1)^6(q^2+q+1)^3, \quad (\text{B.1})$$

$$\begin{aligned} \mathfrak{M}_4(q) &= 16q^{42}(q-1)^{26}(q+1)^6(q^2+1)^8 \\ &\quad \times (9q^4 + 31q^3 + 45q^2 + 31q + 9)^3. \end{aligned} \quad (\text{B.2})$$

For $n = 5$,

$$\begin{aligned} \mathfrak{M}_5(q) &= 256q^{150}(q-1)^{60}(q+1)^{14}(q^2+q+1)^4(q^4+q^3+q^2+q+1)^{15} \\ &\quad \times (36q^6 + 97q^5 + 181q^4 + 213q^3 + 181q^2 + 97q + 36)^3 H_5(q)^2, \end{aligned} \quad (\text{B.3})$$

where

$$H_5(q) = 8q^{10} - 5q^9 - 7q^8 - 6q^7 - 8q^6 + 35q^5 - 8q^4 - 6q^3 - 7q^2 - 5q + 8. \quad (\text{B.4})$$

The reciprocal polynomial H_5 has exactly one zero in $(0, 1)$, namely

$$q_0 = 0.8401533783083766586629364318891329431013925836583 \dots$$

At this base, the critical points in the first and third zero intervals are

$$\begin{aligned} c_0 &= 1.0714044413496100275076056970032887643067724227285 \dots, \\ c_2 &= 1.5701685824623346994394512430684932917838814269399 \dots, \end{aligned}$$

and

$$P_5(c_0; q_0) = P_5(c_2; q_0) = -0.000295418321326036978942747465329242197437155822 \dots$$

Thus $H_5 = 0$ is a genuine Maxwell component, not merely a degeneracy of one critical point.

For $n = 6$, after removal of endpoint, cyclotomic, and manifestly positive factors, one collision factor is the reciprocal degree-forty polynomial

$$\begin{aligned}
H_6(q) = & 200q^{40} + 1387q^{39} + 4911q^{38} + 11453q^{37} + 18354q^{36} + 17819q^{35} + 69q^{34} \\
& - 36849q^{33} - 76446q^{32} - 85091q^{31} - 33909q^{30} + 68239q^{29} + 160240q^{28} \\
& + 159280q^{27} + 27236q^{26} - 177962q^{25} - 321574q^{24} - 286466q^{23} - 74232q^{22} \\
& + 176918q^{21} + 287794q^{20} + 176918q^{19} - 74232q^{18} - 286466q^{17} - 321574q^{16} \\
& - 177962q^{15} + 27236q^{14} + 159280q^{13} + 160240q^{12} + 68239q^{11} - 33909q^{10} \\
& - 85091q^9 - 76446q^8 - 36849q^7 + 69q^6 + 17819q^5 + 18354q^4 + 11453q^3 \\
& + 4911q^2 + 1387q + 200.
\end{aligned} \tag{B.5}$$

High-precision computation finds a positive zero

$$q = 0.7773703152388489309\dots,$$

and continuation numerically identifies it with a collision of the critical values from the first and third zero intervals; this sheet identification is not interval-certified. The complete exact factorization of $\mathfrak{M}_6$, including very large positive factors, is recorded in `data/secondary_discriminants.txt`; keeping it as machine-readable text avoids several pages of typographically uninformative coefficients.

Remark B.1. Reciprocity of the collision factors is consistent with the reversal identity

$$P_n(a; q) = (-a)^n q^{n(n-1)/2} P_n(a^{-1} q^{1-n}; q),$$

together with the transformation $q \mapsto q^{-1}$. It does not by itself identify which pair of critical sheets collides. That labeling requires continuation of the real critical points or interval-certified root isolation.

C Reproducible numerical and symbolic experiments

C.1 Archive layout

The archive contains the following source and generated artifacts.

Path	Purpose
<code>inverse_q_analogs_report.tex</code>	Complete LaTeX source of this report.
<code>inverse_q_analogs_report.pdf</code>	Compiled report.
<code>numerical_experiments.py</code>	Documented Python program that regenerates every table, exact discriminant factorization, and figure used in the report.
<code>requirements.txt</code>	Minimal Python package requirements.
<code>data/finite_critical_values.csv</code>	Critical points and critical values of finite products for selected (n, q) .
<code>data/infinite_critical_asymptotics.csv</code>	Exact numerical critical points and heights compared with their remote asymptotic formulas.
<code>data/principal_inverse_derivative_signs.csv</code>	Derivatives of $\mathcal{A}_q$ through order eight at a grid of test points.
<code>data/qbinomial_inverse_accuracy.csv</code>	Exact Gaussian-coefficient inverses near $q = 1$, compared with linear, quadratic, and cubic reversion.

Path	Purpose
data/qgamma_minima.csv	The minimizer x_q^* , minimum value, and curvature of Γ_q .
data/secondary_discriminants.txt	Exact symbolic factorizations of $\mathfrak{M}_n$ for $3 \leq n \leq 6$.
figures/*.pdf	Vector figures generated by the Python program and included in the report.

C.2 Reproduction command

From the report directory, run

```
python3 numerical_experiments.py
latexmk -pdf -interaction=nonstopmode inverse_q_analogs_report.tex
```

The numerical program uses high-precision arithmetic for branch points and root solving, exact integer polynomial arithmetic for discriminants, and ordinary double precision only for plotting after the high-precision data have been generated. It fixes no random seed because no randomized algorithm is used.

C.3 Checks implemented in the program

The script performs the following independent consistency checks.

- (i) Every finite critical point is bracketed between consecutive zeros and satisfies the logarithmic-derivative equation to the working precision.
- (ii) At the $n = 5$ collision, two separately continued critical points have critical values agreeing to approximately fifty decimal digits.
- (iii) Remote infinite-product critical points are compared with both the location and height asymptotics, and the ratios approach one.
- (iv) Gaussian inverse values are substituted into the exact finite product formula before the asymptotic errors are recorded.
- (v) Derivatives of the principal product inverse are generated from a local power-series reversion rather than finite differences.
- (vi) The q -gamma minimizer is obtained from the zero of ψ_q , and positive q -trigamma verifies that the stationary point is a strict minimum.

These checks do not replace a proof of conjectural global statements, but they make every numerical claim in the report directly auditable.

D Notation and branch conventions

Symbol	Meaning
$(a; q)_n$	Finite q -Pochhammer product $\prod_{j=0}^{n-1} (1 - aq^j)$.
$(a; q)_\infty$	Infinite product for $ q < 1$.
$(a; q)_\alpha$	Complex-order continuation $(a; q)_\infty / (aq^\alpha; q)_\infty$, with a fixed branch of q^α .

Symbol	Meaning
$P_n(a; q)$	The finite direct product $(a; q)_n$.
$P(a; q)$	The infinite direct product $(a; q)_\infty$.
$\mathcal{A}_q(y)$	Principal real inverse in the argument: $(\mathcal{A}_q(y); q)_\infty = y$, $\mathcal{A}_q(y) < 1$, for $0 < q < 1$, $y > 0$.
$\mathcal{Q}_{n,k}(y)$	Unique positive real base satisfying $[n]_q = y$, for $0 < k < n$ and $y > 1$.
$[x]_q$	q -number $(1 - q^x)/(1 - q)$, with a branch specified for complex powers.
Γ_q, ψ_q	q -gamma and q -digamma functions.
B_q	q -beta function.
$D_n(y; q)$	Primary discriminant $\text{Disc}_a(P_n(a; q) - y)$.
$\mathfrak{M}_n(q)$	Secondary discriminant $\text{Disc}_y D_n(y; q)$.
Caustic	Parameter values where a critical point is degenerate or a direct map loses ordinary fold structure.
Maxwell locus	Parameter values where distinct critical points have the same critical value.
Principal branch	The real branch selected by a monotone interval and, in complex continuation, by a stated base point and path.

Unless stated otherwise, real-product results use $0 < q < 1$. Complex powers use a locally fixed logarithm, $q^z = \exp(z \text{Log } q)$. Consequently, inverses in an exponent or order naturally repeat on a logarithmic lattice. Square roots in Puiseux expansions are fixed by continuation from the real side or by an explicitly chosen local sheet.

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